United States Patent	12407479
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Huang; Yi et al.

Techniques to enhance HARQ-ACK multiplexing on PUSCH with repetitions

Abstract

Apparatus, methods, and computer-readable media for facilitating HARQ feedback multiplexing on PUSCH with repetitions are disclosed herein. An example method for wireless communication at a UE includes receiving an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator. The example method also includes transmitting a first repetition of the uplink transmission. The first repetition may include multiplexing of a quantity of HARQ-ACK feedback bits HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, and the quantity of the HARQ-ACK feedback bits may be based on the bit indicator. The first repetition and the HARQ-ACK feedback may overlap in a time domain.

Inventors: **Huang; Yi (San Diego, CA), Rungta; Pranay Sudeep (New York, NY), Gaal; Peter (San Diego, CA), Yang; Wei (San Diego, CA)**

Applicant: **QUALCOMM Incorporated** (San Diego, CA)

Family ID: **88414866**

Assignee: **QUALCOMM Incorporated (San Diego, CA)**

Appl. No.: **18/051491**

Filed: **October 31, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230344604 A1	Oct. 26, 2023

Related U.S. Application Data

us-provisional-application US 63363424 20220422

Publication Classification

Int. Cl.: **H04L5/00** (20060101); **H04L1/1812** (20230101); **H04W72/1268** (20230101); **H04W72/23** (20230101)

U.S. Cl.:

CPC **H04L5/0055** (20130101); **H04L1/1812** (20130101); **H04W72/1268** (20130101); **H04W72/23** (20230101);

Field of Classification Search

CPC: H04L (5/0055); H04L (1/1812); H04W (72/1268); H04W (72/23)

References Cited

OTHER PUBLICATIONS

Huawei et al: "Enhancements on the scheduling of PUSCH over multiple slots", 3GPP Draft; R1-2103745, 3rd Generation Partnership Project (3GPP), Mobile Competence Centre; 650, Route Des Lucioles; F-06921 Sophia-Antipolis Cedex; France vol. RAN WG1, No. E-meeting; Apr. 12, 2021-Apr. 20, 2021 Apr. 7, 2021 . (Year: 2021). cited by examiner

CATT: "Discussion on HARQ-ACK multiplexing in PUSCH without PUCCH", 3GPP Draft; R1-2201324, 3rd Generation Partnership Project (3GPP), Mobile Competence Centre; 650, Route Des Lucioles; F-06921 Sophia-Antipolis Cedex; France vol. RAN WG1, No. e-Meeting; Feb. 21, 2022-Mar. 3, 2022 Feb. 14, 2022 (Year: 2022). cited by examiner

CATT: "Discussion on HARQ-ACK Multiplexing in PUSCH without PUCCH", 3GPP TSG RAN WG1 #108-e, R1-2201324, 3rd Generation Partnership Project (3GPP), Mobile Competence Centre, 650, Route Des Lucioles, F-06921 Sophia-Antipolis Cedex, France, vol. RAN WG1, No. e-Meeting, Feb. 21, 2022-Mar. 3, 2022, 4 Pages, Feb. 14, 2022 (Feb. 14, 2022), XP052109383, pp. 1-3. cited by applicant

Huawei, et al., "Enhancements on the Scheduling of PUSCH Over Multiple Slots", 3GPP TSG RAN WG1 Meeting #104bis-e, R1-2103745, 3rd Generation Partnership Project (3GPP), Mobile Competence Centre, 650, Route Des Lucioles, F-06921 Sophia-Antipolis Cedex, France, vol. RAN WG1, No. E-meeting, Apr. 12, 2021-Apr. 20, 2021, 6 Pages, Apr. 7, 2021 (Apr. 7, 2021), XP052178367, pp. 3-5. cited by applicant

International Search Report and Written Opinion—PCT/US2023/018875—ISA/EPO—Jun. 15, 2023. cited by applicant

Moderator (NTT Docomo), et al., "Summary on Rel-17 TEI Related Discussion", 3GPP TSG RAN WG1 #104bis-e, R1-2103604, 3rd Generation Partnership Project (3GPP), Mobile Competence Centre, 650, Route Des Lucioles, F-06921 Sophia-Antipolis Cedex, France, vol. RAN WG1, No. e-Meeting, Apr. 12, 2021-Apr. 20, 2021, 37 Pages, Apr. 23, 2021 (Apr. 23, 2021), XP051998105, pp. 26-30. cited by applicant

---

Primary Examiner: Pham; Brenda H

Attorney, Agent or Firm: Procopio, Cory, Hargreaves & Savitch LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) (1) This application claims the benefit of and priority to U.S. Provisional Application Ser. No. 63/363,424, entitled "TECHNIQUES TO ENHANCE HARQ-ACK MULTIPLEXING ON PUSCH WITH REPETITIONS," and filed on Apr. 22, 2022, which is expressly incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to communication systems, and more particularly, to wireless communications utilizing uplink transmission grants with repetitions.

INTRODUCTION

(2) Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources. Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, and time division synchronous code division multiple access (TD-SCDMA) systems.

(3) These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level. An example telecommunication standard is 5G New Radio (NR). 5G NR is part of a continuous mobile broadband evolution promulgated by Third Generation Partnership Project (3GPP) to meet new requirements associated with latency, reliability, security, scalability (e.g., with Internet of Things (IoT)), and other requirements. 5G NR includes services associated with enhanced mobile broadband (eMBB), massive machine type communications (mMTC), and ultra-reliable low latency communications (URLLC). Some aspects of 5G NR may be based on the 4G Long Term Evolution (LTE) standard. There exists a need for further improvements in 5G NR technology. These improvements may also be applicable to other multi-access technologies and the telecommunication standards that employ these technologies.

BRIEF SUMMARY

(4) The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects. This summary neither identifies key or critical elements of all aspects nor delineates the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

(5) In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided for wireless communication. An apparatus may include a user equipment (UE). The example apparatus may receive an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator. The example apparatus may also transmit a first repetition of the uplink transmission, the first repetition multiplexing a quantity of Hybrid Automatic Repeat Request (HARD) acknowledgement (HARQ-ACK) feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator.

(6) In another aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided for wireless communication. An apparatus may include a network entity, such as a base station. The example apparatus output an uplink grant scheduling an uplink transmission

associated with a repetition quantity at a UE, the uplink grant including a bit indicator. The example apparatus may also obtain a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator.
(7) To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram illustrating an example of a wireless communications system and an access network.
(2) FIG. **2**A is a diagram illustrating an example of a first frame, in accordance with various aspects of the present disclosure.
(3) FIG. **2**B is a diagram illustrating an example of downlink (DL) channels within a subframe, in accordance with various aspects of the present disclosure.
(4) FIG. **2**C is a diagram illustrating an example of a second frame, in accordance with various aspects of the present disclosure.
(5) FIG. **2**D is a diagram illustrating an example of uplink (UL) channels within a subframe, in accordance with various aspects of the present disclosure.
(6) FIG. **3** is a diagram illustrating an example of a base station and user equipment (UE) in an access network.
(7) FIG. **4** illustrates an example communication flow between a base station and a UE, in accordance with the teachings disclosed herein.
(8) FIG. **5** is a diagram illustrating a timeline of a UE performing hybrid automatic retransmission (HARQ) feedback multiplexing on a physical uplink shared channel (PUSCH) with repetitions, in accordance with the teachings disclosed herein.
(9) FIG. **6**A is a diagram illustrating a timeline of a UE performing HARQ feedback multiplexing on PUSCH with repetitions based on scheduling information from a base station, in accordance with the teachings disclosed herein.
(10) FIG. **6**B is a diagram illustrating a timeline of the UE performing HARQ feedback multiplexing on PUSCH with repetitions based on scheduling information from the base station, in accordance with the teachings disclosed herein.
(11) FIG. **7** is a diagram illustrating a timeline of a UE performing HARQ feedback multiplexing on PUSCH with repetitions based on scheduling information from a base station, in accordance with the teachings disclosed herein.
(12) FIG. **8** is a diagram illustrating a timeline of a UE performing HARQ feedback multiplexing on PUSCH with repetitions based on scheduling information from a base station, in accordance with the teachings disclosed herein.
(13) FIG. **9** illustrates an example communication flow between a base station and a UE, in accordance with the teachings disclosed herein.
(14) FIG. **10** is a diagram illustrating a timeline of a UE performing HARQ feedback multiplexing on PUSCH with repetitions based on scheduling information from a base station, in accordance with the teachings disclosed herein.
(15) FIG. **11** is a flowchart of a method of wireless communication at a UE, in accordance with the

teachings disclosed herein.
(16) FIG. **12** is a flowchart of a method of wireless communication at a UE, in accordance with the teachings disclosed herein.
(17) FIG. **13** is a diagram illustrating an example of a hardware implementation for an example network entity.
(18) FIG. **14** is a flowchart of a method of wireless communication at a network entity, in accordance with the teachings disclosed herein.
(19) FIG. **15** is a flowchart of a method of wireless communication at a network entity, in accordance with the teachings disclosed herein.
(20) FIG. **16** is a diagram illustrating an example of a hardware implementation for an example network entity.
DETAILED DESCRIPTION
(21) A wireless communication system may support data transmission with HARQ, for example, to improve reliability. For HARQ, a transmitter device may send an initial transmission of a message and may send one or more additional transmissions of the message, if needed, until a termination event occurs. For example, the termination event may include an indication that the message is decoded correctly by a receiver device, or a maximum quantity of transmissions of the message has occurred. After each transmission of the message, the receiver device may send an acknowledgement (ACK) if the message is decoded correctly, or a negative acknowledgement (NACK) if the message is decoded in error or missed. The transmitter device may send another transmission of the message if a NACK is received and may terminate transmission of the message if an ACK is received. As used herein, a “message” may also be referred to as a transport block, a packet, a codeword, a data block, etc.
(22) In some examples, the transmitter device may send the one or more transmissions of the message based on scheduling information. For example, a transmitter device, such as a UE, may receive an uplink grant scheduling the UE to transmit an uplink message, such as on a PUSCH. Additionally, or alternatively, the receiver device may receive the one or more transmissions of a message based on scheduling information. For example, a receiver device, such as a UE, may receive a downlink grant scheduling the UE to receive a downlink message, such as on a PDSCH.
(23) In some scenarios, a device may be configured to transmit a first message and to receive a second message. For example, a UE may receive a downlink grant scheduling the UE to receive a downlink message and may also receive an uplink grant scheduling the UE to transmit one or more transmissions of an uplink message. In some such examples, a repetition of the uplink message may overlap in time with transmission of the HARQ feedback associated with the downlink message.
(24) Aspects disclosed herein provide techniques for improving HARQ feedback multiplexing on PUSCH with repetitions. The disclosed techniques may provide higher data rates, improve capacity, and/or improve spectral efficiency. For example, aspects disclosed herein may configure a UE to determine to perform multiplexing on a repetition regardless of whether the repetition overlaps in time with HARQ feedback. Additionally, the UE may perform the multiplexing based on a quantity of bits indicated by an uplink grant. By determining to perform the multiplexing based on the quantity of bits indicated by an uplink grant (e.g., a bit indicator) and regardless of whether there is an overlap in time, the UE may resolve concerns associated with missed downlink grants. In some examples, the bit indicator may include a bit indicator set including a number of bit indicators. For example, the quantity of bit indicators in the bit indicator set may correspond to a quantity of repetitions configured for an uplink message. The quantity of repetitions may be indicated via the uplink grant and/or via radio resource control (RRC) signaling. In some such examples, the UE may use a respective bit indicator of the bit indicator set to determine a quantity of z bits to multiplex on a respective repetition. In other examples, the bit indicator set may include one bit indicator that is applied to each repetition of the uplink message.

(25) In another aspect, the UE may be configured to determine whether to perform multiplexing on a repetition based on a bit indicator included in an uplink grant. For example, when the value of the bit indicator satisfies a threshold (e.g., x bits of the bit indicator is greater than or equal to the threshold), the UE may perform multiplexing on a repetition regardless of whether the repetition overlaps in time with HARQ feedback scheduled by a downlink grant. In examples in which the value of the bit indicator does not satisfy the threshold (e.g., the x bits is less than the threshold), then the UE may multiplex the x bits of the HARQ feedback on a repetition when the repetition overlaps in time with the HARQ feedback. The UE may also skip multiplexing or multiplex zero bits on a repetition when the repetition is non-overlapping in time with the HARQ feedback.
(26) The detailed description set forth below in connection with the drawings describes various configurations and does not represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.
(27) Several aspects of telecommunication systems are presented with reference to various apparatus and methods. These apparatus and methods are described in the following detailed description and illustrated in the accompanying drawings by various blocks, components, circuits, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using electronic hardware, computer software, or any combination thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.
(28) By way of example, an element, or any portion of an element, or any combination of elements may be implemented as a “processing system” that includes one or more processors. Examples of processors include microprocessors, microcontrollers, graphics processing units (GPUs), central processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems on a chip (SoC), baseband processors, field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. One or more processors in the processing system may execute software. Software, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise, shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software components, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, or any combination thereof.
(29) Accordingly, in one or more example aspects, implementations, and/or use cases, the functions described may be implemented in hardware, software, or any combination thereof. If implemented in software, the functions may be stored on or encoded as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer storage media. Storage media may be any available media that can be accessed by a computer. By way of example, such computer-readable media can comprise a random-access memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer.
(30) While aspects, implementations, and/or use cases are described in this application by illustration to some examples, additional or different aspects, implementations and/or use cases may come about in many different arrangements and scenarios. Aspects, implementations, and/or use cases described herein may be implemented across many differing platform types, devices, systems, shapes, sizes, and packaging arrangements. For example, aspects, implementations, and/or

use cases may come about via integrated chip implementations and other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, artificial intelligence (AI)-enabled devices, etc.). While some examples may or may not be specifically directed to use cases or applications, a wide assortment of applicability of described examples may occur. Aspects, implementations, and/or use cases may range a spectrum from chip-level or modular components to non-modular, non-chip-level implementations and further to aggregate, distributed, or original equipment manufacturer (OEM) devices or systems incorporating one or more techniques herein. In some practical settings, devices incorporating described aspects and features may also include additional components and features for implementation and practice of claimed and described aspect. For example, transmission and reception of wireless signals necessarily includes a number of components for analog and digital purposes (e.g., hardware components including antenna, RF-chains, power amplifiers, modulators, buffer, processor(s), interleaver, adders/summers, etc.). Techniques described herein may be practiced in a wide variety of devices, chip-level components, systems, distributed arrangements, aggregated or disaggregated components, end-user devices, etc. of varying sizes, shapes, and constitution.
(31) Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core network node, a network element, or a network equipment, such as a base station (BS), or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a BS (such as a Node B (NB), evolved NB (eNB), NR BS, 5G NB, access point (AP), a transmission reception point (TRP), or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.
(32) An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).
(33) Base station operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.
(34) FIG. **1** is a diagram **100** illustrating an example of a wireless communications system and an access network. The illustrated wireless communications system includes a disaggregated base station architecture. The disaggregated base station architecture may include one or more CUs (e.g., a CU **110**) that can communicate directly with a core network **120** via a backhaul link, or indirectly with the core network **120** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (MC) (e.g., a Near-RT MC **125**) via an E2

link, or a Non-Real Time (Non-RT) MC **115** associated with a Service Management and Orchestration (SMO) Framework (e.g., an SMO Framework **105**), or both). A CU **110** may communicate with one or more DUs (e.g., a DU **130**) via respective midhaul links, such as an F1 interface. The DU **130** may communicate with one or more RUs (e.g., an RU **140**) via respective fronthaul links. The RU **140** may communicate with respective UEs (e.g., a UE **104**) via one or more radio frequency (RF) access links. In some implementations, the UE **104** may be simultaneously served by multiple RUs.

(35) Each of the units, i.e., the CUs (e.g., a CU **110**), the DUs (e.g., a DU **130**), the RUs (e.g., an RU **140**), as well as the Near-RT RICs (e.g., the Near-RT MC **125**), the Non-RT RICs (e.g., the Non-RT MC **115**), and the SMO Framework **105**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or to transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or to transmit signals over a wired transmission medium to one or more of the other units.

(36) Additionally, the units can include a wireless interface, which may include a receiver, a transmitter, or a transceiver (such as an RF transceiver), configured to receive or to transmit signals, or both, over a wireless transmission medium to one or more of the other units.

(37) In some aspects, the CU **110** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **110**. The CU **110** may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane functionality (i.e., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **110** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as an E1 interface when implemented in an O-RAN configuration. The CU **110** can be implemented to communicate with the DU **130**, as necessary, for network control and signaling.

(38) The DU **130** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs. In some aspects, the DU **130** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation, demodulation, or the like) depending, at least in part, on a functional split, such as those defined by 3GPP. In some aspects, the DU **130** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **130**, or with the control functions hosted by the CU **110**.

(39) Lower-layer functionality can be implemented by one or more RUs. In some deployments, an RU **140**, controlled by a DU **130**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU **140** can be implemented to handle over the air (OTA) communication with one or more UEs (e.g., the UE **104**). In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU **140** can be controlled by a corresponding DU. In some scenarios, this configuration can enable the DU(s) and the CU **110** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

(40) The SMO Framework **105** may be configured to support RAN deployment and provisioning of

non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **105** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements that may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **105** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **190**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs, DUs, RUs and Near-RT RICs. In some implementations, the SMO Framework **105** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **111**, via an O1 interface. Additionally, in some implementations, the SMO Framework **105** can communicate directly with one or more RUs via an O1 interface. The SMO Framework **105** also may include a Non-RT RIC **115** configured to support functionality of the SMO Framework **105**.

(41) The Non-RT RIC **115** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, artificial intelligence (AI)/machine learning (ML) (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **125**. The Non-RT RIC **115** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **125**. The Near-RT RIC **125** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs, one or more DUs, or both, as well as an O-eNB, with the Near-RT MC **125**.

(42) In some implementations, to generate AI/ML models to be deployed in the Near-RT MC **125**, the Non-RT RIC **115** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **125** and may be received at the SMO Framework **105** or the Non-RT RIC **115** from non-network data sources or from network functions. In some examples, the Non-RT RIC **115** or the Near-RT RIC **125** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **115** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **105** (such as reconfiguration via **01**) or via creation of RAN management policies (such as A1 policies).

(43) At least one of the CU **110**, the DU **130**, and the RU **140** may be referred to as a base station **102**. Accordingly, a base station **102** may include one or more of the CU **110**, the DU **130**, and the RU **140** (each component indicated with dotted lines to signify that each component may or may not be included in the base station **102**). The base station **102** provides an access point to the core network **120** for a UE **104**. The base station **102** may include macrocells (high power cellular base station) and/or small cells (low power cellular base station). The small cells include femtocells, picocells, and microcells. A network that includes both small cell and macrocells may be known as a heterogeneous network. A heterogeneous network may also include Home Evolved Node Bs (eNBs) (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG). The communication links between the RUs (e.g., the RU **140**) and the UEs (e.g., the UE **104**) may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to an RU **140** and/or downlink (DL) (also referred to as forward link) transmissions from an RU **140** to a UE **104**. The communication links may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links may be through one or more carriers. The base station **102**/UE **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100, 400, etc. MHz) bandwidth per carrier allocated in a carrier aggregation of up to a total of Yx MHz (x component carriers) used for transmission in each direction. The carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than

for UL). The component carriers may include a primary component carrier and one or more secondary component carriers. A primary component carrier may be referred to as a primary cell (PCell) and a secondary component carrier may be referred to as a secondary cell (SCell).
(44) Certain UEs may communicate with each other using device-to-device (D2D) communication (e.g., a D2D communication link **158**). The D2D communication link **158** may use the DL/UL wireless wide area network (WWAN) spectrum. The D2D communication link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and a physical sidelink control channel (PSCCH). D2D communication may be through a variety of wireless D2D communications systems, such as for example, Bluetooth, Wi-Fi based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standard, LTE, or NR.
(45) The wireless communications system may further include a Wi-Fi AP **150** in communication with a UE **104** (also referred to as Wi-Fi stations (STAs)) via communication link **154**, e.g., in a 5 GHz unlicensed frequency spectrum or the like. When communicating in an unlicensed frequency spectrum, the UE **104**/Wi-Fi AP **150** may perform a clear channel assessment (CCA) prior to communicating in order to determine whether the channel is available.
(46) The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). Although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.
(47) The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR2-2 (52.6 GHz-71 GHz), FR4 (71 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.
(48) With the above aspects in mind, unless specifically stated otherwise, the term “sub-6 GHz” or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, the term “millimeter wave” or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR2-2, and/or FR5, or may be within the EHF band.
(49) The base station **102** and the UE **104** may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate beamforming. The base station **102** may transmit a beamformed signal **182** to the UE **104** in one or more transmit directions. The UE **104** may receive the beamformed signal from the base station **102** in one or more receive directions. The UE **104** may also transmit a beamformed signal **184** to the base station **102** in one or more transmit directions. The base station **102** may receive the beamformed signal from the UE **104** in one or more receive directions. The base station **102**/UE **104** may perform beam training to determine the best receive and transmit directions for each of the base station **102**/UE **104**. The transmit and receive directions for the base station **102** may or may not be the same. The transmit and receive directions for the UE **104** may or may not be the same.

(50) The base station **102** may include and/or be referred to as a gNB, Node B, eNB, an access point, a base transceiver station, a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a transmission reception point (TRP), network node, network entity, network equipment, or some other suitable terminology. The base station **102** can be implemented as an integrated access and backhaul (IAB) node, a relay node, a sidelink node, an aggregated (monolithic) base station with a baseband unit (BBU) (including a CU and a DU) and an RU, or as a disaggregated base station including one or more of a CU, a DU, and/or an RU. The set of base stations, which may include disaggregated base stations and/or aggregated base stations, may be referred to as next generation (NG) RAN (NG-RAN).

(51) The core network **120** may include an Access and Mobility Management Function (AMF) (e.g., an AMF **161**), a Session Management Function (SMF) (e.g., an SMF **162**), a User Plane Function (UPF) (e.g., a UPF **163**), a Unified Data Management (UDM) (e.g., a UDM **164**), one or more location servers **168**, and other functional entities. The AMF **161** is the control node that processes the signaling between the UE **104** and the core network **120**. The AMF **161** supports registration management, connection management, mobility management, and other functions. The SMF **162** supports session management and other functions. The UPF **163** supports packet routing, packet forwarding, and other functions. The UDM **164** supports the generation of authentication and key agreement (AKA) credentials, user identification handling, access authorization, and subscription management. The one or more location servers **168** are illustrated as including a Gateway Mobile Location Center (GMLC) (e.g., a GMLC **165**) and a Location Management Function (LMF) (e.g., an LMF **166**). However, generally, the one or more location servers **168** may include one or more location/positioning servers, which may include one or more of the GMLC **165**, the LMF **166**, a position determination entity (PDE), a serving mobile location center (SMLC), a mobile positioning center (MPC), or the like. The GMLC **165** and the LMF **166** support UE location services. The GMLC **165** provides an interface for clients/applications (e.g., emergency services) for accessing UE positioning information. The LMF **166** receives measurements and assistance information from the NG-RAN and the UE **104** via the AMF **161** to compute the position of the UE **104**. The NG-RAN may utilize one or more positioning methods in order to determine the position of the UE **104**. Positioning the UE **104** may involve signal measurements, a position estimate, and an optional velocity computation based on the measurements. The signal measurements may be made by the UE **104** and/or the serving base station (e.g., the base station **102**). The signals measured may be based on one or more of a satellite positioning system (SPS) **170** (e.g., one or more of a Global Navigation Satellite System (GNSS), global position system (GPS), non-terrestrial network (NTN), or other satellite position/location system), LTE signals, wireless local area network (WLAN) signals, Bluetooth signals, a terrestrial beacon system (TB S), sensor-based information (e.g., barometric pressure sensor, motion sensor), NR enhanced cell ID (NR E-CID) methods, NR signals (e.g., multi-round trip time (Multi-RTT), DL angle-of-departure (DL-AoD), DL time difference of arrival (DL-TDOA), UL time difference of arrival (UL-TDOA), and UL angle-of-arrival (UL-AoA) positioning), and/or other systems/signals/sensors.

(52) Examples of UEs include a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a global positioning system, a multimedia device, a video device, a digital audio player (e.g., MP3 player), a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an electric meter, a gas pump, a large or small kitchen appliance, a healthcare device, an implant, a sensor/actuator, a display, or any other similar functioning device. Some of the UEs may be referred to as IoT devices (e.g., parking meter, gas pump, toaster, vehicles, heart monitor, etc.). The UE **104** may also be referred to as a station, a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a

remote terminal, a handset, a user agent, a mobile client, a client, or some other suitable terminology. In some scenarios, the term UE may also apply to one or more companion devices such as in a device constellation arrangement. One or more of these devices may collectively access the network and/or individually access the network.
(53) Referring again to FIG. **1**, in certain aspects, a device in communication with a network entity, such as a UE **104** in communication with a base station **102** or a component of a base station (e.g., a CU **110**, a DU **130**, and/or an RU **140**), may be configured to manage one or more aspects of wireless communication. For example, the UE **104** may include a repetition component **198** configured to facilitate performing HARQ feedback multiplexing on PUSCH with repetitions.
(54) In certain aspects, the repetition component **198** may be configured to receive an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator. The example repetition component **198** may also be configured to transmit a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator.
(55) In another configuration, a network entity, such as a base station **102** or a component of a base station (e.g., a CU **110**, a DU **130**, and/or an RU **140**), may be configured to manage or more aspects of wireless communication. For example, the base station **102** may include a scheduling component **199** configured to facilitate performing HARQ feedback multiplexing on PUSCH with repetitions.
(56) In certain aspects, the scheduling component **199** may be configured to output an uplink grant scheduling an uplink transmission associated with a repetition quantity at a UE, the uplink grant including a bit indicator. The example scheduling component **199** may also be configured to obtain a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator.
(57) Aspects disclosed herein provide techniques for improving HARQ feedback multiplexing on PUSCH with repetitions. The disclosed techniques may provide higher data rates, improve capacity and/or improve spectral efficiency.
(58) Although the following description provides examples directed to 5G NR (and, in particular, to single uplink transmission grants with repetitions), the concepts described herein may be applicable to other similar areas, such as LTE, LTE-A, CDMA, GSM, and/or other wireless technologies, in which a UE may multiplex HARQ feedback on PUSCH with repetitions.
(59) FIG. **2**A is a diagram **200** illustrating an example of a first subframe within a 5G NR frame structure. FIG. **2**B is a diagram **230** illustrating an example of DL channels within a 5G NR subframe. FIG. **2**C is a diagram **250** illustrating an example of a second subframe within a 5G NR frame structure. FIG. **2**D is a diagram **280** illustrating an example of UL channels within a 5G NR subframe. The 5G NR frame structure may be frequency division duplexed (FDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for either DL or UL, or may be time division duplexed (TDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for both DL and UL. In the examples provided by FIGS. **2**A, **2**C, the 5G NR frame structure is assumed to be TDD, with subframe 4 being configured with slot format 28 (with mostly DL), where D is DL, U is UL, and F is flexible for use between DL/UL, and subframe 3 being configured with slot format 1 (with all UL). While subframes 3, 4 are shown with slot formats 1, 28, respectively, any particular subframe may be configured with any of the various available slot formats 0-61. Slot formats 0, 1 are all DL, UL, respectively. Other slot formats 2-61 include a mix of DL, UL, and flexible symbols. UEs are configured with the slot format (dynamically through DL

control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling) through a received slot format indicator (SFI). Note that the description infra applies also to a 5G NR frame structure that is TDD.

(60) FIGS. **2**A-**2**D illustrate a frame structure, and the aspects of the present disclosure may be applicable to other wireless communication technologies, which may have a different frame structure and/or different channels. A frame (10 ms) may be divided into 10 equally sized subframes (1 ms). Each subframe may include one or more time slots. Subframes may also include mini-slots, which may include 7, 4, or 2 symbols. Each slot may include 14 or 12 symbols, depending on whether the cyclic prefix (CP) is normal or extended. For normal CP, each slot may include 14 symbols, and for extended CP, each slot may include 12 symbols. The symbols on DL may be CP orthogonal frequency division multiplexing (OFDM) (CP-OFDM) symbols. The symbols on UL may be CP-OFDM symbols (for high throughput scenarios) or discrete Fourier transform (DFT) spread OFDM (DFT-s-OFDM) symbols (for power limited scenarios; limited to a single stream transmission). The number of slots within a subframe is based on the CP and the numerology. The numerology defines the subcarrier spacing (SCS) (see Table 1). The symbol length/duration may scale with 1/SCS.

(61) TABLE-US-00001 TABLE 1 Numerology, SCS, and CP SCS μ Δf = 2.sup.μ .Math. 15[kHz] Cyclic prefix 0 15 Normal 1 30 Normal 2 60 Normal, Extended 3 120 Normal 4 240 Normal 5 480 Normal 6 960 Normal

(62) For normal CP (14 symbols/slot), different numerologies μ 0 to 4 allow for 1, 2, 4, 8, and 16 slots, respectively, per subframe. For extended CP, the numerology 2 allows for 4 slots per subframe. Accordingly, for normal CP and numerology μ, there are 14 symbols/slot and 2.sup.μ slots/subframe. As shown in Table 1, the subcarrier spacing may be equal to 2.sup.μ*15 kHz, where μ is the numerology 0 to 4. As such, the numerology μ=0 has a subcarrier spacing of 15 kHz and the numerology μ=4 has a subcarrier spacing of 240 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. **2**A-**2**D provide an example of normal CP with 14 symbols per slot and numerology μ=2 with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 μs. Within a set of frames, there may be one or more different bandwidth parts (BWPs) (see FIG. **2**B) that are frequency division multiplexed. Each BWP may have a particular numerology and CP (normal or extended).

(63) A resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.

(64) As illustrated in FIG. **2**A, some of the REs carry reference (pilot) signals (RS) for the UE. The RS may include demodulation RS (DM-RS) (indicated as R for one particular configuration, but other DM-RS configurations are possible) and channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and phase tracking RS (PT-RS).

(65) FIG. **2**B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs) (e.g., 1, 2, 4, 8, or 16 CCEs), each CCE including six RE groups (REGs), each REG including 12 consecutive REs in an OFDM symbol of an RB. A PDCCH within one BWP may be referred to as a control resource set (CORESET). A UE is configured to monitor PDCCH candidates in a PDCCH search space (e.g., common search space, UE-specific search space) during PDCCH monitoring occasions on the CORESET, where the PDCCH candidates have different DCI formats and different aggregation levels. Additional BWPs may be located at greater and/or lower frequencies across the channel bandwidth. A primary synchronization signal (PSS) may be within symbol 2 of particular subframes of a frame. The PSS is used by a UE **104** to determine subframe/symbol timing and a physical layer identity. A secondary synchronization signal (SSS)

may be within symbol 4 of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing. Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the DM-RS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block (also referred to as SS block (SSB)). The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIBs), and paging messages.

(66) As illustrated in FIG. **2**C, some of the REs carry DM-RS (indicated as R for one particular configuration, but other DM-RS configurations are possible) for channel estimation at the base station. The UE may transmit DM-RS for the physical uplink control channel (PUCCH) and DM-RS for the physical uplink shared channel (PUSCH). The PUSCH DM-RS may be transmitted in the first one or two symbols of the PUSCH. The PUCCH DM-RS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. The UE may transmit sounding reference signals (SRS). The SRS may be transmitted in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.

(67) FIG. **2**D illustrates an example of various UL channels within a subframe of a frame. The PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and hybrid automatic repeat request (HARQ) acknowledgment (ACK) (HARQ-ACK) feedback (i.e., one or more HARQ ACK bits indicating one or more ACK and/or negative ACK (NACK)). The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.

(68) FIG. **3** is a block diagram that illustrates an example of a first wireless device that is configured to exchange wireless communication with a second wireless device. In the illustrated example of FIG. **3**, the first wireless device may include a base station **310**, the second wireless device may include a UE **350**, and the base station **310** may be in communication with the UE **350** in an access network. As shown in FIG. **3**, the base station **310** includes a transmit processor (TX processor **316**), a transmitter **318**Tx, a receiver **318**Rx, antennas **320**, a receive processor (RX processor **370**), a channel estimator **374**, a controller/processor **375**, and memory **376**. The example UE **350** includes antennas **352**, a transmitter **354**Tx, a receiver **354**Rx, an RX processor **356**, a channel estimator **358**, a controller/processor **359**, memory **360**, and a TX processor **368**. In other examples, the base station **310** and/or the UE **350** may include additional or alternative components.

(69) In the DL, Internet protocol (IP) packets may be provided to the controller/processor **375**. The controller/processor **375** implements layer 3 and layer 2 functionality. Layer 3 includes a radio resource control (RRC) layer, and layer 2 includes a service data adaptation protocol (SDAP) layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The controller/processor **375** provides RRC layer functionality associated with broadcasting of system information (e.g., MIB, SIBs), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter radio access technology (RAT) mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer packet data units (PDUs), error correction through ARQ, concatenation,

segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

(70) The TX processor **316** and the RX processor **370** implement layer 1 functionality associated with various signal processing functions. Layer 1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The TX processor **316** handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an OFDM subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an Inverse Fast Fourier Transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM stream is spatially precoded to produce multiple spatial streams. Channel estimates from the channel estimator **374** may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE **350**. Each spatial stream may then be provided to a different antenna of the antennas **320** via a separate transmitter (e.g., the transmitter **318**Tx). Each transmitter **318**Tx may modulate a radio frequency (RF) carrier with a respective spatial stream for transmission.

(71) At the UE **350**, each receiver **354**Rx receives a signal through its respective antenna of the antennas **352**. Each receiver **354**Rx recovers information modulated onto an RF carrier and provides the information to the RX processor **356**. The TX processor **368** and the RX processor **356** implement layer 1 functionality associated with various signal processing functions. The RX processor **356** may perform spatial processing on the information to recover any spatial streams destined for the UE **350**. If multiple spatial streams are destined for the UE **350**, two or more of the multiple spatial streams may be combined by the RX processor **356** into a single OFDM symbol stream. The RX processor **356** then converts the OFDM symbol stream from the time-domain to the frequency domain using a Fast Fourier Transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station **310**. These soft decisions may be based on channel estimates computed by the channel estimator **358**. The soft decisions are then decoded and deinterleaved to recover the data and control signals that were originally transmitted by the base station **310** on the physical channel. The data and control signals are then provided to the controller/processor **359**, which implements layer 3 and layer 2 functionality.

(72) The controller/processor **359** can be associated with the memory **360** that stores program codes and data. The memory **360** may be referred to as a computer-readable medium. In the UL, the controller/processor **359** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets. The controller/processor **359** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

(73) Similar to the functionality described in connection with the DL transmission by the base station **310**, the controller/processor **359** provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the

transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto TBs, demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

(74) Channel estimates derived by the channel estimator **358** from a reference signal or feedback transmitted by the base station **310** may be used by the TX processor **368** to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the TX processor **368** may be provided to different antenna of the antennas **352** via separate transmitters (e.g., the transmitter **354**Tx). Each transmitter **354**Tx may modulate an RF carrier with a respective spatial stream for transmission.

(75) The UL transmission is processed at the base station **310** in a manner similar to that described in connection with the receiver function at the UE **350**. Each receiver **318**Rx receives a signal through its respective antenna of the antennas **320**. Each receiver **318**Rx recovers information modulated onto an RF carrier and provides the information to the RX processor **370**.

(76) The controller/processor **375** can be associated with the memory **376** that stores program codes and data. The memory **376** may be referred to as a computer-readable medium. In the UL, the controller/processor **375** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets. The controller/processor **375** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

(77) At least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359** may be configured to perform aspects in connection with the repetition component **198** of FIG. **1**.

(78) At least one of the TX processor **316**, the RX processor **370**, and the controller/processor **375** may be configured to perform aspects in connection with the scheduling component **199** of FIG. **1**.

(79) A wireless communication system may support data transmission with HARQ, for example, to improve reliability. For HARQ, a transmitter device may send an initial transmission of a message and may send one or more additional transmissions of the message, if needed, until a termination event occurs. For example, the termination event may include an indication that the message is decoded correctly by a receiver device, or a maximum quantity of transmissions of the message has occurred. After each transmission of the message, the receiver may send an ACK, sometimes referred to as a HARQ-ACK, if the message is decoded correctly, or a NACK, sometimes referred to as a HARQ-NACK, if the message is decoded in error or missed. The transmitter device may send another transmission of the message if a NACK is received and may terminate transmission of the message if an ACK is received.

(80) In some examples, the transmitter device may send the one or more transmissions of the message based on scheduling information. For example, a transmitter device, such as a UE, may receive an uplink grant scheduling the UE to transmit an uplink message, such as on a PUSCH. Additionally, or alternatively, the receiver device may receive the one or more transmissions of a message based on scheduling information. For example, a receiver device, such as a UE, may receive a downlink grant scheduling the UE to receive a downlink message, such as on a PDSCH.

(81) In some scenarios, a device may be configured to transmit a first message and to receive a second message. For example, a UE may receive a downlink grant scheduling the UE to receive a downlink message and an uplink grant scheduling the UE to transmit one or more transmissions of an uplink message. In some such examples, a repetition of the uplink message may overlap in time with transmission of HARQ feedback associated with the downlink message.

(82) FIG. **4** illustrates an example communication flow **400** between a network entity **402** and a UE **404**, as presented herein. One or more aspects described for the network entity **402** may be performed by a component of a base station, such as a CU, a DU, and/or an RU. In the illustrated

example, the communication flow **400** facilitates the UE **404** performing HARQ feedback multiplexing on PUSCH with repetitions. Aspects of the network entity **402** may be implemented by the base station **102** of FIG. **1** and/or the base station **310** of FIG. **3**. Aspects of the UE **404** may be implemented by the UE **104** of FIG. **1** and/or the UE **350** of FIG. **3**. Although not shown in the illustrated example of FIG. **4**, in additional or alternative examples, the network entity **402** and/or the UE **404** may be in communication with one or more other base stations or UEs.

(83) As shown in FIG. **4**, the network entity **402** may transmit a downlink grant **410** that is received by the UE **404**. The network entity **402** may transmit the downlink grant **410** on a PDCCH. The downlink grant **410** may include information related to a downlink message **420**, such as PDSCH resources **412** allocated for monitoring occasions associated with one or more transmissions of a downlink message **420**. The downlink grant **410** may also include information related to HARQ feedback **432** (e.g., an ACK or a NACK) for the downlink message **420**. For example, the downlink grant **410** may indicate HARQ resources **414** allocated to the UE **404** for transmitting the HARQ feedback **432**, and include a downlink bit indicator **416** indicating a quantity of ACK/NACK bits allocated for the HARQ feedback **432**. In some aspects, the downlink bit indicator **416** may include a downlink assignment index (DAI) that may be used to facilitate determination of the quantity of bits for the HARQ feedback **432**.

(84) The network entity **402** may transmit the downlink message **420** that is received by the UE **404**. The network entity **402** may transmit the downlink message **420** on a PDSCH. The UE **404** may receive the downlink message **420** based on the PDSCH resources **412** indicated by the downlink grant **410**. For example, the PDSCH resources **412** may indicate a first monitoring occasion **422** and the UE **404** may monitor resources associated with the first monitoring occasion **422** to receive the downlink message **420**. The first monitoring occasion **422** may be associated with time resources and/or frequency resources.

(85) In the example of FIG. **4**, the UE **404** may perform a generation procedure **430** to generate the HARQ feedback **432** based on the downlink message **420**. For example, the UE **404** may generate uplink control information (UCI) with an ACK or a NACK based on if the UE **404** decodes the downlink message **420** at the first monitoring occasion **422**.

(86) As shown in FIG. **4**, the UE **404** may transmit the HARQ feedback **432** associated with the downlink message **420**. The UE **404** may use resources allocated by the downlink grant **410** for the HARQ feedback **432**. For example, the HARQ resources **414** may indicate time resources and/or frequency resources associated with a HARQ occasion **434**. The UE **404** may transmit the HARQ feedback **432** based on the HARQ occasion **434**. Additionally, a payload size of the HARQ feedback **432** may be based on the downlink bit indicator **416** of the downlink grant **410**. The UE **404** may transmit the HARQ feedback **432** using UCI on a physical uplink control channel (PUCCH).

(87) In the example of FIG. **4**, the UE **404** may also be configured to transmit an uplink message that is received by the network entity **402**. For example, the network entity **402** may transmit an uplink grant **440** that is received by the UE **404**. The network entity **402** may transmit the uplink grant **440** on a PDCCH. The uplink grant **440** may include information related to the uplink message, such as PUSCH resources **442** (e.g., time resources and/or frequency resources) allocated to the UE **404** for transmission of the uplink message. The uplink grant **440** may also include a repetition indicator **444** indicating a quantity of repetitions of the uplink message. For example, based on the uplink grant **440**, the UE **404** may determine to transmit k repetitions of the uplink message, may determine a first resource allocated to the initial transmission of the uplink message, and may determine subsequent resources allocated for the k−1 repetitions of the uplink message. As described above, the UE **404** may transmit up to k repetitions of the uplink message until a termination event occurs.

(88) Although the example of FIG. **4** includes the repetition indicator **444** with the uplink grant **440**, in other examples, the UE **404** may receive the repetition indicator **444** via RRC signaling. For

example, the network entity **402** may transmit RRC signaling **436** that is received by the UE **404**. The RRC signaling **436** may include the repetition indicator **444** indicating a quantity of repetitions of the uplink message.

(89) In some examples, a repetition of the uplink message may coincide (e.g., overlap) in time with transmission of HARQ feedback. In examples in which an uplink message repetition and HARQ feedback overlap, the UE **404** may multiplex the HARQ feedback with the repetition. The UE **404** may perform a determination procedure **450** to determine whether UCI carrying the HARQ feedback **432** overlaps with a repetition of the uplink message. For example, the UE **404** may determine that a first repetition **452** of the uplink message is non-overlapping with the HARQ occasion **434**. In such examples, the UE **404** may transmit the first repetition **452** of the uplink message that is received by the network entity **402**. The UE **404** may transmit the first repetition **452** while skipping multiplexing on the first repetition **452** or by multiplexing zero bits on the first repetition **452**. The UE **404** may transmit the first repetition **452** on a PUSCH.

(90) In examples in which an uplink message repetition overlaps with HARQ feedback, the UE **404** may multiplex the HARQ feedback and the repetition. For example, when performing the determination procedure **450**, the UE **404** may determine that a second repetition **456** is overlapping in time with the HARQ occasion **434**. In such examples, the UE **404** may perform a multiplexing procedure **454** to multiplex the HARQ feedback **432** with the repetition (e.g., the second repetition **456**).

(91) In some examples, the UE **404** may multiplex x bits of the HARQ feedback **432** with the second repetition **456**. The UE **404** may receive a quantity of bits of the HARQ feedback **432** to multiplex with the second repetition **456** via the uplink grant **440**. For example, the uplink grant **440** may include a bit indicator **446** indicating a quantity of bits (e.g., x bits) to multiplex. In such examples, when the UE **404** determines that an uplink message repetition and HARQ feedback are overlapping in time, the UE **404** may multiplex the x bits, indicated by the bit indicator **446**, of the HARQ feedback **432** with the second repetition **456**. In some aspects, the bit indicator **446** may correspond to a total DAI (TDAI) that may be used to facilitate determination of the quantity of bits of the HARQ feedback **432** to multiplex. The TDAI may be referred to as a “UL-TDAI” or by any other name.

(92) In examples in which the bit indicator **446** indicates a quantity of bits that is less than or equal to a threshold, the UE **404** may puncture x bits of the repetition. In examples in which the bit indicator **446** indicates a quantity of bits that is greater than the threshold, the UE **404** may multiplex the HARQ feedback **432** by rate-matching the respective repetition of the uplink message.

(93) As shown in FIG. **4**, the UE **404** may transmit the second repetition **456** of the uplink message that is received by the network entity **402**. The UE **404** may transmit the second repetition **456** on a PUSCH. The second repetition **456** may be a repetition of the uplink message multiplexed by x bits of the HARQ feedback **432**.

(94) FIG. **5** is a diagram illustrating a timeline **500** of a UE **504** performing HARQ feedback multiplexing on PUSCH with repetitions, as presented herein. In the illustrated example of FIG. **5**, the UE **504** may receive scheduling information for a downlink message and for an uplink message from a network entity **502**. Aspects of the network entity **502** may be implemented by the network entity **402** of FIG. **4**. Aspects of the UE **504** may be implemented by the UE **404** of FIG. **4**.

(95) In the illustrated example of FIG. **5**, the UE **504** may receive a downlink grant **510** from the network entity **502** at time T1. Aspects of the downlink grant **510** may be implemented by the downlink grant **410** of FIG. **4**. The downlink grant **510** may provide information for receiving a PDSCH **512**. Aspects of the PDSCH **512** may be implemented by the downlink message **420** of FIG. **4**. The downlink grant **510** may include PDSCH resources **510***a*, such as the PDSCH resources **412**, that indicate a monitoring occasion associated with time T2. The downlink grant **510** may also indicate HARQ resources **510***b*, such as the HARQ resources **414**, that indicate a

HARQ occasion for the UE **504** to transmit HARQ feedback **514** associated with the PDSCH **512**. For example, the downlink grant **510** may include HARQ resources indicating a HARQ occasion at time T6. The HARQ feedback **514** may include an ACK when the UE **504** decodes the PDSCH **512** and may otherwise include a NACK.

(96) As shown in FIG. **5**, the UE **504** may receive an uplink grant **520** from the network entity **502** at time T3. Aspects of the uplink grant **520** may be implemented by the uplink grant **440** of FIG. **4**. For example, the uplink grant **520** may indicate PUSCH resources **520***a*, such as the PUSCH resources **442**, a repetition indicator **520***b*, such as the repetition indicator **444**, and a bit indicator **520***c*, such as the bit indicator **446**. In the example of FIG. **5**, the repetition indicator **520***b* may indicate that the UE **504** is to transmit four repetitions of the uplink message. For example, and based on the PUSCH resources **520***a* of the uplink grant **520**, the UE **504** may determine to transmit a first repetition **522** ("PUSCH Repetition 1") at time T4, transmit a second repetition **524** ("PUSCH Repetition 2") at time T5, transmit a third repetition **526** ("PUSCH Repetition 3") at time T6, and transmit a fourth repetition **528** ("PUSCH Repetition 4") at time T7.

(97) In the example of FIG. **5**, the UE **504** may use the bit indicator **520***c* of an uplink grant **520** to multiplex x bits on a repetition when HARQ feedback scheduled by a downlink grant overlaps with the respective repetition. For example, based on the HARQ occasion associated with the HARQ feedback **514** and the PUSCH resources **520***a* associated with the repetitions of the uplink message, the UE **504** may determine that the first repetition **522**, the second repetition **524**, and the fourth repetition **528** are non-overlapping in time with the HARQ occasion. In such examples, the UE **504** may transmit the respective repetitions without multiplexing, as described in connection with the determination procedure **450** and the first repetition **452** of FIG. **4**.

(98) In examples in which the UE **504** determines that a HARQ occasion and a repetition are overlapping in time (e.g., the third repetition **526** and the HARQ occasion associated with the HARQ feedback **514** at time T6), the UE **504** may multiplex x bits of the HARQ feedback **514** on the respective repetition, as described in connection with the multiplexing procedure **454** and the second repetition **456** of FIG. **4**. For example, the uplink grant **520** includes the bit indicator **520***c* ("TDAI") indicating a quantity of x bits to multiplex. In such examples, the UE **504** may multiplex x bits of the HARQ feedback **514** with the third repetition **526**. As described above, multiplexing the x bits of the HARQ feedback **514** with the third repetition **526** may include puncturing x bits of the third repetition **526** or rate-matching the third repetition **526**.

(99) The network entity **502** may receive the repetitions of the uplink message based on expected multiplexing of bits for each repetition. For example, in the example of FIG. **5**, the UE **504** may multiplex {0, 0, x, 0} bits for the respective repetitions of the uplink message. Since the network entity **502** may schedule the HARQ occasion associated with the HARQ feedback **514** and the repetitions of the uplink message, the network entity **502** may perform demultiplexing of each repetition based on a determination that the first repetition **522**, the second repetition **524**, and the fourth repetition **528** are non-overlapping with a HARQ occasion and that the third repetition **526** is overlapping with the HARQ occasion. That is, based on the scheduled transmissions of the repetitions and the HARQ occasion, the network entity **502** may determine that the third repetition **526** is a multiplexed repetition and that the remaining repetitions are non-multiplexed repetitions. Additionally, since the network entity **502** provides the bit indicator **520***c* with the uplink grant **520** (e.g., the TDAI x bits), the network entity **502** has the ability to determine that x bits of the HARQ feedback **514** are multiplexed with the third repetition **526**. That is, to receive the repetitions of the uplink message, the network entity **502** may apply {0, 0, x, 0} demultiplexing bits for the respective repetitions of the uplink message.

(100) In the example of FIG. **5**, the UE **504** is configured with one HARQ occasion at time T6 and is provided one bit indicator (e.g., the bit indicator **520***c* of the uplink grant **520**) that the UE **504** uses to determine how many bits of the HARQ feedback **514** to multiplex when the HARQ occasion overlaps in time with a repetition. In some examples, the UE may be configured with

multiple HARQ occasions for different respective downlink grants. Additionally, one or more of the HARQ occasions may overlap in time with respective repetitions of an uplink message.

(101) FIG. **6**A and FIG. **6**B are diagrams illustrating respective timelines of performing HARQ feedback multiplexing on PUSCH with repetitions.

(102) FIG. **6**A is a diagram illustrating a timeline **600** of a UE **604** performing HARQ feedback multiplexing on PUSCH with repetitions based on scheduling information from a network entity **602**, as presented herein. In the illustrated example of FIG. **6**A, the UE **604** may receive scheduling information for multiple downlink messages and for an uplink message from the network entity **602**. Aspects of the network entity **602** may be implemented by the network entity **402** of FIG. **4**. Aspects of the UE **604** may be implemented by the UE **404** of FIG. **4**.

(103) In the illustrated example of FIG. **6**A, the network entity **602** may transmit three downlink grants that are received by the UE **604**. The three downlink grants may schedule three downlink messages and three HARQ occasions for transmitting respective HARQ feedback. For example, and as shown in FIG. **6**A, the UE **604** may be configured with a first HARQ occasion at time T2 for transmitting first HARQ feedback **610** ("DL Grant 1 ACK/NACK"), a second HARQ occasion at time T4 for transmitting second HARQ feedback **612** ("DL Grant 2 ACK/NACK"), and a third HARQ occasion at time T5 for transmitting third HARQ feedback **614** ("DL Grant 3 ACK/NACK"). Although not shown in the example of FIG. **6**A, it may be appreciated that the first HARQ feedback **610** is associated with a first downlink message at a first monitoring occasion, as described in connection with the HARQ feedback **514** and the PDSCH **512** of FIG. **5**. In a similar manner, the second HARQ feedback **612** may be associated with a second downlink message at a second monitoring occasion and the third HARQ feedback **614** may be associated with a third downlink message at a third monitoring occasion.

(104) Similar to the example of FIG. **5**, the UE **604** of FIG. **6**A may receive an uplink grant **620** from the network entity **602** at time T1. In the illustrated example of FIG. **6**A, the UE **604** determines to schedule a quantity of repetitions of an uplink message at respective times based on the repetition indicator and the PUSCH resources of the uplink grant **620**. For example, the UE **604** may schedule four repetitions of an uplink message. In the example of FIG. **6**A, the UE **604** may schedule a first repetition **622** ("PUSCH Repetition 1") at time T2, a second repetition **624** ("PUSCH Repetition 2") at time T3, a third repetition **626** ("PUSCH Repetition 3") at time T4, and a fourth repetition **628** ("PUSCH Repetition 4") at time T5 based on the uplink grant **620**.

(105) In the example of FIG. **6**A, the uplink grant **620** includes a bit indicator **621** ("TDAI") indicating to multiplex 2 bits when an overlap in time occurs between a HARQ occasion scheduled by a downlink grant and a repetition. Additionally, each of the downlink grants associated with the first HARQ feedback **610**, the second HARQ feedback **612**, and the third HARQ feedback **614** includes a respective downlink bit indicator indicating a payload size of each respective HARQ feedback. For example, the payload size of the first HARQ feedback **610** is one bit, the payload size of the second HARQ feedback **612** is two bits, and the payload size of the third HARQ feedback **614** is three bits.

(106) In the example of FIG. **6**A, when a PUSCH repetition is non-overlapping with HARQ feedback scheduled by a downlink grant, the UE **604** disregards the bit indicator **621** of the uplink grant **620** and multiplexes 0 bits on the PUSCH repetition. For example, the UE **604** may multiplex 0 bits on the second repetition **624** at time T3.

(107) In examples in which a PUSCH repetition is overlapping with HARQ feedback scheduled by a downlink grant, the UE **604** may multiplex z bits on the PUSCH repetition. The value of the z bits may be determined based on a relationship between the bit indicator **621** of the uplink grant **620** (e.g., the x bits) and the downlink bit indicator of each downlink grant associated with respective HARQ feedback (e.g., the y bit payload size of each HARQ feedback).

(108) For example, the UE **604** may use Equation 1 (below) to determine the z bits.

$z=4T+x$ bits, where $T=0, 1, 2, \ldots$ Equation 1:

(109) In Equation 1, the term z represents the quantity of bits to multiplex on the PUSCH repetition, the term x represents the value of the bit indicator **621**, and the term T represents a modifier. The value of the modifier T may be determined as the smallest integer that satisfies Equation 2 (below).

ceil($4T+x \geq y$), where T=0, 1, 2, . . . Equation 2:

(110) In Equation 2, the term x represents the value of the bit indicator **621**, the term y represents the payload size of the HARQ feedback, and the term T represents a modifier. The UE **604** may determine the value of the modifier T as the smallest integer that satisfies Equation 2. In examples in which the value of the bit indicator **621** (e.g., x bits) is greater than or equal to the value of the downlink bit indicator (e.g., y bits) of a HARQ feedback (e.g., x bits≥y bits), the value of the modifier Tis zero and then Equation 1 may be simplified and the UE **604** may use Equation 3 (below) to determine value of the z bits.

$z=x$ bits Equation 3:

(111) That is, when the value of the bit indicator **621** (e.g., x bits) is greater than or equal to the value of the downlink bit indicator (e.g., y bits) of a HARQ feedback (e.g., x bits≥y bits), the UE **604** uses the x bits indicated by the bit indicator **621** to multiplex on the PUSCH repetition. For example, with respect to the first repetition **622** and the first HARQ feedback **610**, the 2 bits indicated by the bit indicator **621** is greater than the 1 bit payload size of the first HARQ feedback **610** and, thus, the UE **604** multiplexes two bits on the first repetition **622** at time T2. In a similar manner, and with respect to the third repetition **626** and the second HARQ feedback **612**, the 2 bits indicated by the bit indicator **621** is equal to the 2 bits payload size of the second HARQ feedback **612** and, thus, the UE **604** multiplexes two bits on the second repetition **624** at time T4. It may be appreciated that when the value of the bit indicator **621** is greater than the value of the downlink bit indicator of a HARQ feedback, the UE **604** may use one or more dummy bits when multiplexing, such as adding one dummy bit when multiplexing on the first repetition **622** at time T2. Examples of a dummy bit include a NACK indicator.

(112) In examples in which the value of the bit indicator **621** is less than the value of the downlink bit indicator of a HARQ feedback (e.g., x bits<y bits), then the UE **604** first uses Equation 2 (above) to determine the value of the term T and then uses Equation 1 (above) to determine the value of the z bits. For example, and with respect to the fourth repetition **628** (e.g., x is 2 bits) and the third HARQ feedback **614** (e.g., y payload size is 3 bits) at time T5, the value of the modifier Tis one.

(113) ceil($4T + (2) \geq (3)$) Equation2 ceil($4T \geq 1$)ceil($T \geq \frac{1}{4}$)$T = 1$

(114) Using the value of the modifier T as one, the UE **604** may use Equation 1 to determine the value of the z bits. For example, using Equation 1, the UE **604** may determine the value of z bits is 6 bits.

z=4(1)+(2)

z=6 Equation 1:

(115) That is, with respect to the transmission at time T5, the UE **604** may multiplex six bits on the fourth repetition **628**. Similar to the example of the transmission at time T2, the UE **604** may add dummy bits when the z bits is greater than the payload size of the respective HARQ feedback. For example, the UE **604** may add three dummy bits (e.g., three NACK indicators) to the three bits of the third HARQ feedback **614** to satisfy multiplexing the six bits on the fourth repetition **628**.

(116) Thus, as shown in FIG. **6**A, the UE **604** may multiplex {2, 0, 2, 6} bits for the repetitions of the uplink message at times T2, T3, T4, and T5, respectively. Additionally, the network entity **602** may expect the repetitions of the uplink message to be multiplexed with expected multiplexing bits {2, 0, 2, 6} bits. The network entity **602** may use the expected multiplexing bits to demultiplex the respective repetitions, as described in connection with the example of FIG. **5**.

(117) FIG. **6**B is a diagram illustrating a timeline **650** of the UE **604** performing HARQ feedback

multiplexing on PUSCH with repetitions based on scheduling information from the network entity **602**, as presented herein. Similar to the example of FIG. **6**A, the network entity **602** may transmit scheduling information for multiple downlink messages and for an uplink message to the UE **604**. For example, the network entity **602** may transmit an uplink grant **660** that is received by the UE **604** at time T3. Based on the uplink grant **660**, the UE **604** may schedule four repetitions of an uplink message. For example, the UE **604** may schedule a first repetition **662** ("PUSCH Repetition 1") at time T4, a second repetition **664** ("PUSCH Repetition 2") at time T5, a third repetition **666** ("PUSCH Repetition 3") at time T6, and a fourth repetition **668** ("PUSCH Repetition 4") at time T7.

(118) In the example of FIG. **6**B, the network entity **602** transmits a first downlink grant **670** and a third downlink grant **674** at time T1. The first downlink grant **670** schedules first HARQ feedback **672** at time T4 and the third downlink grant **674** schedules third HARQ feedback **676** at time T7. The network entity **602** also transmits a second downlink grant **680** at time T2. The second downlink grant **680** schedules second HARQ feedback **682** at time T7. Although the example of FIG. **6**B illustrates the first downlink grant **670** and the third downlink grant **674** being transmitted at a same time (e.g., at time T1), in other examples, the first downlink grant **670** and the third downlink grant **674** may be transmitted by the network entity **602** at different times.

(119) In the example of FIG. **6**B, the UE **604** receives the second downlink grant **680** at time T2 and, thus, is scheduled to transmit the second HARQ feedback **682** at time T6. However, in the example of FIG. **6**B, the UE **604** does not receive the first downlink grant **670** and the third downlink grant **674** at time T1 and, thus, is not scheduled to transmit the first HARQ feedback **672** at time T4 and the third HARQ feedback **676** at time T7, respectively.

(120) Similar to the example of FIG. **6**A, the uplink grant **660** includes a bit indicator **661** indicating 2 bits (e.g., TDAI is 2 bits), and the payload size associated with the HARQ feedbacks of FIG. **6**B are the same as the respective HARQ feedbacks of FIG. **6**A. That is, the payload size of the first HARQ feedback **672** is one bit, the payload size of the second HARQ feedback **682** is two bits, and the payload size of the third HARQ feedback **676** is three bits.

(121) As described above, when a repetition is non-overlapping in time with a HARQ feedback scheduled by a downlink grant, the UE transmits the repetition without multiplexing or by multiplexing zero bits. For example, with respect to time T5, the UE **604** may determine that there is no overlap between the second repetition **664** and HARQ feedback and, thus, may transmit the second repetition **664** without multiplexing or by multiplexing zero bits. Additionally, the network entity **602** expects no multiplexing at time T5 with the second repetition **664** and, thus, expects zero multiplexing bits at time T5.

(122) Similar to the example of FIG. **6**A, the UE **604** of FIG. **6**B may determine to multiplex two bits for the transmission at time T6 based on the second HARQ feedback **682** overlapping in time with the third repetition **666**. Additionally, the network entity **602** expects the transmission at time T6 to be multiplexed with two bits.

(123) As shown in FIG. **6**B, the UE **604** does not receive the first downlink grant **670** and the third downlink grant **674**. The UE **604** may not receive a downlink grant, for example, due to bad channel conditions or the downlink channel associated with the downlink grant being an unreliable channel. In such scenarios, the UE **604** is also not scheduled with the HARQ occasions associated with the first HARQ feedback **672** and the third HARQ feedback **676**. For example, since the UE **604** does not receive the first downlink grant **670**, the UE **604** does not receive the PDSCH resources associated with the corresponding downlink message and also does not receive the HARQ resources associated with the first HARQ feedback **672**.

(124) Accordingly, the UE **604** determines that the first repetition **662** and the fourth repetition **668** are non-overlapping in time with HARQ feedback scheduled by downlink grants. Thus, the UE **604** may transmit the first repetition **662** at time T4 and the fourth repetition **668** at time T7 without multiplexing or by multiplexing zero bits.

(125) However, in contrast to the example of the second repetition **664** at time T5, the network entity **602** is expecting an overlap in time between the first repetition **662** and the first HARQ feedback **672** at time T4 and an overlap in time between the fourth repetition **668** and the third HARQ feedback **676** at time T7. Thus, the network entity **602** expects the transmission at time T4 to be multiplexed with two bits and the transmission at time T7 to be multiplexed with six bits, as described in connection with the transmission at time T2 and the transmission at time T5, respectively, of FIG. **6**A.
(126) As shown in the examples of FIG. **6**A and FIG. **6**B, the UE **604** may determine whether to perform multiplexing or skip multiplexing on a repetition based on successfully receiving a downlink grant scheduling HARQ feedback. For example, in the example of FIG. **6**B, the UE **604** misses the first downlink grant **670** and the third downlink grant **674** and, thus, determines not to perform multiplexing on the repetitions at time T4 and time T7. However, the network entity **602** may be unaware that the UE **604** missed the respective downlink grants and expects the respective repetitions to be multiplexed. For example, in the example of FIG. **6**B, the UE **604** may multiplex {0, 0, 2, 0} bits for the repetitions of the uplink message at times T4, T5, T6, and T7, respectively. Additionally, the network entity **602** may expect the repetitions of the uplink message to be multiplexed with expected multiplexing bits {2, 0, 2, 6} bits. When the multiplexed bits and the expected multiplexed bits are not the same, as shown in FIG. **6**B, the network entity **602** may be unable to decode the repetitions and HARQ feedback for the misaligned transmissions.
(127) Additionally, in examples in which the UE **604** determines to perform multiplexing on a repetition, the UE **604** may receive a common bit indicator in the uplink grant that may be applied to each repetition. For example, in the example of FIG. **6**A, the UE **604** uses the value of the bit indicator **621** to determine how many bits to multiplex on the first repetition **622** at time T2, the third repetition **626** at time T4, and the fourth repetition **628** at time T5. However, in some such examples, when determining the z bits to multiplex, the UE **604** may determine to add one or more dummy bits for multiplexing. For example, with respect to the transmission at time T5 of FIG. **6**A, the UE **604** determines to multiplex six bits, but the payload size of the third HARQ feedback **614** is three bits and, thus, the UE **604** adds three dummy bits.
(128) Aspects disclosed herein provide techniques for improving HARQ feedback multiplexing on PUSCH with repetitions. The disclosed techniques may provide higher data rates, improve capacity and/or improve spectral efficiency. For example, aspects disclosed herein may configure the UE to determine to perform multiplexing on a repetition regardless of whether the repetition overlaps in time with HARQ feedback. Additionally, the UE may perform the multiplexing based on a quantity of bits indicated by an uplink grant. By determining to perform the multiplexing based on the quantity of bits indicated by an uplink grant (e.g., the bit indicator) and regardless of whether there is an overlap in time, the UE may resolve concerns associated with missed downlink grants, as described in connection with the example of FIG. **6**B. In some examples, the bit indicator may include a bit indicator set including a number of bit indicators. For example, the quantity of bit indicators in the bit indicator set may correspond to the quantity of repetitions configured for an uplink message. The quantity of repetitions may be indicated via the uplink grant and/or via RRC signaling. In some such examples, the UE may use a respective bit indicator of the bit indicator set to determine a quantity of z bits to multiplex on a respective repetition. In other examples, the bit indicator set may include one bit indicator that is applied to each repetition of the uplink message.
(129) In another aspect, the UE may be configured to determine whether to perform multiplexing on a repetition based on a bit indicator included in an uplink grant. For example, when the value of the bit indicator satisfies a threshold (e.g., the x bits indicated by the bit indicator is greater than or equal to the threshold), the UE may perform multiplexing on a repetition regardless of whether the repetition overlaps in time with HARQ feedback scheduled by a downlink grant. In examples in which the value of the bit indicator does not satisfy the threshold (e.g., the x bits is less than the threshold), then the UE may multiplex the x bits of the HARQ feedback on a repetition when the

repetition overlaps in time with the HARQ feedback. The UE may also skip multiplexing or multiplex zero bits on a repetition when the repetition is non-overlapping in time with the HARQ feedback.

(130) FIG. **7** is a diagram illustrating a timeline **700** of a UE **704** performing HARQ feedback multiplexing on PUSCH with repetitions based on scheduling information from a network entity **702**, as presented herein. One or more aspects described for the network entity **702** may be performed by a component of a base station or a network entity, such as a CU, a DU, and/or an RU. Aspects of the network entity **702** may be implemented by the base station **102** of FIG. **1** and/or the base station **310** of FIG. **3**. Aspects of the UE **704** may be implemented by the UE **104** of FIG. **1** and/or the UE **350** of FIG. **3**.

(131) Although not shown in the example of FIG. **7**, the network entity **702** may transmit downlink grants that are received by the UE **704**. The network entity **702** may transmit the downlink grants on PDCCH. The downlink grants may schedule respective HARQ occasions for transmitting HARQ feedback. For example, the UE **704** may be configured to transmit first HARQ feedback **730** ("DL Grant 1 ACK/NACK") with a payload size of one bit at time T2, may be configured to transmit second HARQ feedback **732** ("DL Grant 2 ACK/NACK") with a payload size of two bits at time T3, and may be configured to transmit third HARQ feedback **734** ("DL Grant 3 ACK/NACK") with a payload size of three bits at time T5.

(132) In the example of FIG. **7**, the network entity **702** transmits an uplink grant **710** that is received by the UE **704** at time T1. The uplink grant **710** may include a single uplink transmission grant with repetitions. For example, the uplink grant **710** may schedule k repetitions of an uplink message. In the example of FIG. **7**, the uplink grant **710** includes a repetition indicator **712** indicating to transmit four repetitions of an uplink message. The uplink grant **710** also includes PUSCH resources **714** indicating time resources and/or frequency resources associated with transmitting each of the repetitions of the uplink message. For example, the PUSCH resources **714** may allocate resources to transmit a first repetition **720** ("PUSCH Repetition 1") at time T2, resources to transmit a second repetition **722** ("PUSCH Repetition 2") at time T3, resources to transmit a third repetition **724** ("PUSCH Repetition 3") at time T4, and resources to transmit a fourth repetition **726** ("PUSCH Repetition 4") at time T5.

(133) The uplink grant **710** also includes a bit indicator, such as the bit indicator **446** of FIG. **4**. In the example of FIG. **7**, the bit indicator includes a bit indicator set **716** including a number of bit indicators. For example, the bit indicator set **716** of FIG. **7** includes a number of bit indicators that is the same as the k repetitions indicated by the repetition indicator **712** (e.g., 4 bit indicators). Each of the different bit indicators of the bit indicator set **716** may be associated with a respective repetition. For example, the bit indicator set **716** of FIG. **7** includes a set of four bit indicators {2, 0, 3, 3}, and where a first bit indicator "2" is associated with the first repetition **720**, a second bit indicator "0" is associated with the second repetition **722**, a third bit indicator "3" is associated with the third repetition **724**, and a fourth bit indicator "3" is associated with the fourth repetition **726**. Aspects of the bit indicator may be implemented by an UL-TDAI parameter. In some examples, the UL-TDAI parameter may include N fields corresponding to the number of bit indicators included in the bit indicator set **716**.

(134) In the example of FIG. **7**, the UE **704** is configured to multiplex a bit indicator of the bit indicator set **716** with a respective repetition of the uplink message. The UE **704** multiplexes a quantity of bits indicated by a bit indicator regardless of whether the respective repetition is overlapping in time with HARQ feedback. For example, based on the first bit indicator of the bit indicator set **716**, the UE **704** multiplexes 2 bits of the first HARQ feedback **730** on the first repetition **720**. That is, the UE **704** may transmit a first transmission **740** at time T2 including two bits of the first HARQ feedback **730** multiplexed on the first repetition **720**. In a similar manner, the UE **704** may transmit a second transmission **742** at time T3 that includes zero bits multiplexed with the second repetition **722** based on the second bit indicator "0" of the bit indicator set **716**,

may transmit a third transmission **744** at time T4 that includes three bits of the second HARQ feedback **732** multiplexed with the third repetition **724** based on the third bit indicator "3" of the bit indicator set **716**, and may transmit a fourth transmission **746** at time T5 that includes three bits of the third HARQ feedback **734** multiplexed with the fourth repetition **726** based on the fourth bit indicator "3" of the bit indicator set **716**.
(135) As shown in FIG. **7**, by receiving the bit indicator set **716** with a number of bit indicators corresponding to the k repetitions of the uplink message, the UE **704** may skip determining whether to perform HARQ feedback multiplexing on a PUSCH repetition based on a downlink grant that may or may not be missed. Thus, as shown in FIG. **7**, the UE **704** may multiplex {2, 0, 3, 3} bits for the repetitions of the uplink message at times T2, T3, T4, and T5, respectively. Additionally, the network entity **702** may expect the repetitions of the uplink message are multiplexed with expected multiplexing {2, 0, 3, 3} bits. The network entity **702** may use the respective expected multiplexing bits to demultiplex the corresponding transmissions and to recover the respective repetition and the respective HARQ feedback, if any, as described in connection with the example of FIG. **5**.
(136) In the illustrated example of FIG. **7**, the UE **704** may use the value of the bit indicator to determine the number of bits to multiplex for a respective repetition. However, in examples in which a downlink grant indicates a payload size of corresponding HARQ feedback, the UE **704** may use Equation 1 (above) and Equation 2 (above) to determine the z bits to multiplex. For example, if the fourth bit indicator of the bit indicator set **716** was "2" bits, then the fourth transmission **746** would include six bits of the third HARQ feedback **734** multiplexed with the fourth repetition **726** based on the fourth bit indicator "2" of the bit indicator set **716**, as described in connection with the transmission at time T5 of FIG. **6**A. In such examples, the UE **704** may include one or more dummy bits to satisfy multiplexing the z bits.
(137) It may be appreciated that since the network entity **702** may provide the scheduling information for the downlink grant and the uplink grant, the network entity **702** may configure the bit indicators of the bit indicator set **716** so that each respective bit indicator is at least equal to the payload size of a corresponding HARQ feedback.
(138) The example of FIG. **7** helps alleviate concerns with HARQ feedback multiplexing on PUSCH repetitions when a downlink grant is missed, for example, which may cause a misalignment between the number of bits multiplexed by the UE and the expected number of multiplexed bits multiplexed at the base station. However, the example bit indicator set **716** includes overhead associated with each bit indicator of the bit indicator set **716**. For example, with four repetitions of an uplink message, the uplink grant also provides four bit indicators to use with respective repetitions. That is, the cost of employing the bit indicator set **716** is the increased overhead of the downlink control channel (e.g., DCI on a PDCCH) due to the N bit indicators included in the bit indicator set **716**.
(139) FIG. **8** is a diagram illustrating a timeline **800** of a UE **804** performing HARQ feedback multiplexing on PUSCH with repetitions based on scheduling information from a network entity **802**, as presented herein. One or more aspects described for the network entity **802** may be performed by a component of a base station or a network entity, such as a CU, a DU, and/or an RU. Aspects of the network entity **802** may be implemented by the base station **102** of FIG. **1** and/or the base station **310** of FIG. **3**. Aspects of the UE **804** may be implemented by the UE **104** of FIG. **1** and/or the UE **350** of FIG. **3**. Although not shown in the example of FIG. **8**, the network entity **802** may transmit downlink grants that may be received by the UE **804** or may be missed by the UE **804**, as described in connection with the example of FIG. **6**B. The network entity **802** may transmit the downlink grants on PDCCH. The downlink grants may schedule respective HARQ occasions for transmitting HARQ feedback.
(140) Similar to the example of FIG. **6**B, the UE **804** may miss receiving a first downlink grant, may receive a second downlink grant, and may miss receiving a third downlink grant. Based on the

three downlink grants transmitted by the network entity **802**, the network entity **802** may expect to receive first HARQ feedback **830** ("DL Grant 1 ACK/NACK") with a payload size of one bit at time T2, may expect to receive second HARQ feedback **832** ("DL Grant 2 ACK/NACK") with a payload size of two bits at time T3, and may expect to receive third HARQ feedback **834** ("DL Grant 3 ACK/NACK") with a payload size of three bits at time T5. However, based on the downlink grants received by the UE **804** (e.g., the second downlink grant), the UE **804** may determine a HARQ occasion for transmitting the second HARQ feedback **832** at time T4.

(141) In the example of FIG. **8**, the network entity **802** transmits an uplink grant **810** that is received by the UE **804** at time T1. The uplink grant **810** may include a single uplink transmission grant with repetitions. For example, the uplink grant **810** may schedule k repetitions of an uplink message. In the example of FIG. **8**, the uplink grant **810** includes a repetition indicator **812** indicating a quantity of four repetitions associated with an uplink message. The uplink grant **810** also includes PUSCH resources **814** indicating time resources and/or frequency resources associated with transmitting each of the repetitions of the uplink message. For example, the PUSCH resources **814** may allocate resources to transmit a first repetition **820** ("PUSCH Repetition 1") at time T2, resources to transmit a second repetition **822** ("PUSCH Repetition 2") at time T3, resources to transmit a third repetition **824** ("PUSCH Repetition 3") at time T4, and resources to transmit a fourth repetition **826** ("PUSCH Repetition 4") at time T5.

(142) Although the example of FIG. **8** includes the repetition indicator **812** with the uplink grant **810**, in other examples, the UE **804** may receive the repetition indicator **812** via RRC signaling, such as the example RRC signaling **436** of FIG. **4**.

(143) The uplink grant **710** also includes a bit indicator **816**. In the example of FIG. **8**, the bit indicator **816** is a common bit indicator indicating a quantity of x bits to multiplex on repetitions. In the example of FIG. **8**, the bit indicator **816** indicates to multiplex three bits on repetitions. Aspects of the bit indicator may be implemented by an UL-TDAI parameter or a common UL-TDAI parameter.

(144) In the example of FIG. **8**, the UE **804** is configured to multiplex the x bits on repetitions regardless of whether the repetition is overlapping in time with HARQ feedback. It may be appreciated that by multiplexing the x bits regardless of whether a repetition is overlapping in time with HARQ feedback, the example techniques resolve concerns associated with missed downlink grants.

(145) For example, based on the three bits indicated by the bit indicator **816**, the UE **804** multiplexes three bits on the first repetition **820**. That is, the UE **804** may transmit a first transmission **840** at time T2 including three multiplexed bits on the first repetition **820**. In a similar manner, the UE **804** may transmit a second transmission **842** at time T3 that includes three bits multiplexed with the second repetition **822**, may transmit a third transmission **844** at time T4 that includes three bits of the second HARQ feedback **832** multiplexed with the third repetition **824**, and may transmit a fourth transmission **846** at time T5 that includes three bits multiplexed with the fourth repetition **826**. In examples in which the x bits indicated by the bit indicator **816** is greater than the payload size of a HARQ feedback, the UE **804** may add one or more dummy bits, such as one or more NACK indicators, so that each transmission includes x bits multiplexed on a respective repetition. In examples in which a downlink grant is missed and, thus, the UE **804** is not scheduled with a corresponding HARQ occasion (e.g., at time T2 with the missed HARQ occasion associated with the first HARQ feedback **830** and at time T5 with the missed HARQ occasion associated with the third HARQ feedback **834**), or when a repetition is non-overlapping in time with HARQ feedback (e.g., at time T3), the x bits that the UE **804** multiplexes with each respective repetition may be one or more dummy bits (e.g., one or more NACK indicators).

(146) Thus, as shown in FIG. **8**, the UE **804** may multiplex {3, 3, 3, 3} bits for the repetitions of the uplink message at times T2, T3, T4, and T5, respectively. Additionally, the network entity **802** may expect the repetitions of the uplink message to be multiplexed with expected multiplexing {3, 3, 3,

3} bits. As described above, the network entity **802** may use the respective expected multiplexing bits to demultiplex the corresponding transmissions and to recover the respective repetition and the respective HARQ feedback, if any, as described in connection with the examples of FIG. **5**.
(147) In the illustrated example of FIG. **8**, the UE **804** uses the value of the bit indicator **816** to determine the number of bits to multiplex for a repetition. However, in examples in which a downlink grant indicates a payload size of corresponding HARQ feedback, the UE **804** may use Equation 1 (above) and Equation 2 (above) to determine the z bits to multiplex. For example, if the bit indicator of the uplink grant **810** of FIG. **8** was "2" bits, then the UE **804** may multiplex six bits of the third HARQ feedback **834** with the fourth repetition **826** when transmitting the fourth transmission **846**, as described in connection with the transmission at time T5 of FIG. **6**A. In such examples, the UE **804** may include one or more dummy bits to satisfy multiplexing the z bits.
(148) The example of FIG. **8** improves HARQ feedback multiplexing on PUSCH repetitions when a downlink grant is missed, for example, which may cause a misalignment between the number of bits multiplexed by the UE and the expected number of multiplexed bits multiplexed at the base station. Additionally, the bit indicator **816** is common (e.g., shared) for each repetition, which reduces overhead associated with adding multiple bit indicators to a bit indicator set, as described in connection with the example of FIG. **7**.
(149) FIG. **9** illustrates an example communication flow **900** between a network entity **902** and a UE **904**, as presented herein. One or more aspects described for the network entity **902** may be performed by a component of a base station or a network entity, such as a CU, a DU, and/or an RU. Aspects of the network entity **902** may be implemented by the base station **102** of FIG. **1** and/or the base station **310** of FIG. **3**. Aspects of the UE **904** may be implemented by the UE **104** of FIG. **1** and/or the UE **350** of FIG. **3**. Although not shown in the illustrated example of FIG. **9**, in additional or alternative examples, the network entity **902** and/or the UE **904** may be in communication with one or more other base stations or UEs.
(150) In the illustrated example, the communication flow **900** facilitates the UE **904** performing HARQ feedback multiplexing on PUSCH with repetitions. For example, the UE **904** may be configured to determine whether to perform multiplexing on a repetition based on a bit indicator included in an uplink grant. In some examples, the UE **904** may perform multiplexing on a repetition regardless of whether the repetition overlaps in time with HARQ feedback when the value of the bit indicator satisfies a threshold (e.g., the x bits is greater than or equal to the threshold), In examples in which the value of the bit indicator does not satisfy the threshold (e.g., the x bits is less than the threshold), then the UE may multiplex the x bits on a repetition when the repetition overlaps in time with HARQ feedback. The UE may also skip multiplexing or multiplex zero bits on a repetition when the repetition is non-overlapping in time with HARQ feedback
(151) As shown in FIG. **9**, the network entity **902** may transmit a downlink grant **910** that is received by the UE **904**. The network entity **902** may transmit the downlink grant **910** in DCI on a PDCCH. Aspects of the downlink grant **910** may be similar to the downlink grant **410** of FIG. **4**. For example, the downlink grant **910** may include information related to a downlink message **920** and HARQ feedback **932** associated with the downlink message **920**, such as PDSCH resources **912** allocated for monitoring occasions associated with one or more transmissions of the downlink message **920**, HARQ resources **914** allocated to the UE **904** for transmitting the HARQ feedback **932**, and a downlink bit indicator **916** indicating a quantity of ACK/NACK bits for the HARQ feedback. In some aspects, the downlink bit indicator **916** may include a TDAI that may be used to facilitate determination of the quantity of bits for the HARQ feedback **932**.
(152) The network entity **902** may transmit the downlink message **920** that is received by the UE **904**. The network entity **902** may transmit the downlink message **920** on a PDSCH. Aspects of the downlink message **920** may be similar to the downlink message **420** of FIG. **4**. The UE **904** may receive the downlink message **920** based on the PDSCH resources **912** indicated by the downlink grant **910**. For example, the PDSCH resources **912** may indicate a first monitoring occasion **922**

and the UE **904** may monitor resources associated with the first monitoring occasion **922** to receive the downlink message **920**. The first monitoring occasion **922** may be associated with time resources and/or frequency resources.

(153) The UE **904** may perform a generation procedure **930** to generate the HARQ feedback **932** based on the downlink message **920**. For example, the UE **904** may generate UCI with an ACK or a NACK based on if the UE **904** decoded the downlink message **920** at the first monitoring occasion **922**. Aspects of the HARQ feedback **932** may be similar to the HARQ feedback **432** of FIG. **4**.

(154) As shown in FIG. **9**, the UE **904** may transmit the HARQ feedback **932** associated with the downlink message **920**. The UE **904** may use resources allocated by the downlink grant **910** for the HARQ feedback **932**. For example, the HARQ resources **914** may indicate time resources and/or frequency resources associated with a HARQ occasion **934**. The UE **904** may transmit the HARQ feedback **932** based on the HARQ occasion **934**. Additionally, a payload size of the HARQ feedback **932** may be based on the downlink bit indicator **916** of the downlink grant **910**. The UE **904** may transmit the HARQ feedback **932** using UCI on a PUCCH.

(155) As shown in FIG. **9**, the UE **904** may also be configured to transmit an uplink message that is received by the network entity **902**. For example, the network entity **902** may transmit an uplink grant **940** that is received by the UE **904**. The network entity **902** may transmit the uplink grant **940** on a PDCCH. Aspects of the uplink grant **940** may be similar to the uplink grant **440** of FIG. **4**. For example, the uplink grant **940** may include information related to the uplink message, such as PUSCH resources **942** allocated to the UE **904** for transmission of the uplink message, a repetition indicator **944** indicating a quantity of repetitions of the uplink message, and a bit indicator **946** indicating a quantity of bits (e.g., x bits) to multiplex with a repetition of the uplink message. In some aspects, the bit indicator **946** may correspond to a TDAI that may be used to facilitate determination of the quantity of bits of the HARQ feedback **932** to multiplex. As described above, the UE **904** may transmit up to k repetitions of the uplink message until a termination event occurs.

(156) Although the example of FIG. **9** includes the repetition indicator **944** with the uplink grant **940**, in other examples, the UE **904** may receive the repetition indicator **944** via RRC signaling, such as the example RRC signaling **436** of FIG. **4**.

(157) In an aspect, the UE **904** may improve HARQ feedback multiplexing on PUSCH with repetitions based on a value of the bit indicator **946** of the uplink grant **940**. For example, the UE **904** may determine if the x bits indicated by the bit indicator **946** is greater than a threshold quantity of bits. In some examples, the threshold quantity of bits may be two bits. For example, when the x bits is less than or equal to two bits, then the UE **904** may multiplex the x bits on a repetition by puncturing the x bits of the repetition. The network entity **902** may have the capability to decode a transmission that includes a repetition with zero bits, one bit, or two bits punctured. However, when the x bits is greater than the threshold (e.g., greater than two bits), then the UE **904** multiplexes the x bits by rate-matching the repetition. The network entity **902** may have difficulty attempting to decode a transmission that includes rate-matching when the quantity of multiplexed bits (e.g., at the UE **904**) is different than the quantity of expected multiplexed bits (e.g., at the network entity **902**).

(158) In the example of FIG. **9**, the UE **904** may perform a determination procedure **950** to determine if the x bits indicated by the bit indicator **946** is greater than a threshold quantity of bit (e.g., is x bits>2 bits?). If the UE **904** determines that the x bits is greater than the threshold quantity, then, the UE **904** performs a multiplexing procedure **952** to multiplex z bits on a repetition. The UE **904** may use Equation 1, Equation 2, and/or Equation 3 (reproduced below) to determine the value of the z bits.

$z=4T+x$ bits, where $T=0, 1, 2, \ldots$ Equation 1:

$\mathrm{ceil}(4T+x \geq y)$, where $T=0, 1, 2, \ldots$ Equation 2:

$z=x$ bits Equation 3:

(159) As shown in FIG. **9**, the UE **904** may transmit a first transmission **954** that is received by the

network entity **902**. The UE **904** may transmit the first transmission **954** on a PUSCH. The first transmission **954** may include multiplexing of z bits on a repetition. The UE **904** may multiplex the z bits regardless of whether the repetition overlaps in time with HARQ feedback. By multiplexing the z bits regardless of the overlapping status of the repetition, the multiplexing does not depend on whether HARQ feedback scheduled by a downlink grant is overlapping. Thus, in scenarios in which a downlink grant is missed by the UE **904**, for example, as described in connection with the timeline **650** of FIG. **6**B and/or the timeline **800** of FIG. **8**, the quantity of multiplexed bits (e.g., at the UE **904**) is aligned with the quantity of expected multiplexed bits (e.g., at the network entity **902**) and, thus, the network entity **902** may decode the first transmission **954** to receive the repetition and the HARQ feedback, if any.

(160) If the UE **904** determines, via the determination procedure **950**, that the x bits dose not satisfy the threshold (e.g., the x bits is less than or equal to the threshold quantity of bits), then the UE **904** may perform multiplexing of bits on a repetition based on whether the repetition is overlapping in time with HARQ feedback. For example, the UE **904** may perform a determination procedure **960** to determine whether UCI carrying the HARQ feedback **932** is overlapping in time with a repetition of the uplink message. For example, the UE **904** may determine that a repetition of the uplink message is non-overlapping with the HARQ occasion **934**. In such examples, the UE **904** may transmit a second transmission **962** that is received by the network entity **902**. The UE **904** may transmit the second transmission **962** while skipping multiplexing on the repetition or by multiplexing zero bits on the repetition. The UE **904** may transmit the second transmission **962** on a PUSCH.

(161) In examples in which a repetition overlaps with HARQ feedback, the UE **904** may multiplex the HARQ feedback and the repetition. For example, the UE **904** may determine, via the determination procedure **960**, that a repetition is overlapping in time with the HARQ occasion **934**. In such examples, the UE **904** may perform a multiplexing procedure **964** to multiplex x bits of the HARQ feedback **932** with the repetition. As described above in connection with the determination procedure **950**, the value of the x bits is less than or equal to the threshold quantity of bits (e.g., 2 bits). Thus, the UE **904** may puncture, via the multiplexing procedure **964**, the x bits of the repetition.

(162) As shown in FIG. **9**, the UE **904** may transmit a third transmission **966** that is received by the network entity **902**. The UE **904** may transmit the third transmission **966** on a PUSCH. The third transmission **966** may include x bits of the HARQ feedback **932** multiplexed on a repetition.

(163) FIG. **10** is a diagram illustrating a timeline **1000** of a UE **1004** performing HARQ feedback multiplexing on PUSCH with repetitions based on scheduling information from a network entity **1002**, as presented herein. One or more aspects described for the network entity **1002** may be performed by a component of a base station or a network entity, such as a CU, a DU, and/or an RU. Aspects of the network entity **1002** may be implemented by the base station **102** of FIG. **1**, the base station **310** of FIG. **3**, and/or the network entity **902** of FIG. **9**. Aspects of the UE **1004** may be implemented by the UE **104** of FIG. **1**, the UE **350** of FIG. **3**, and/or the UE **904** of FIG. **9**.

(164) Although not shown in the example of FIG. **10**, the network entity **1002** may transmit downlink grants that may be received by the UE **1004** or may be missed by the UE **1004**, as described in connection with the examples of FIG. **6**B and FIG. **8**. The network entity **1002** may transmit the downlink grants on PDCCH. The downlink grants may schedule respective HARQ occasions for transmitting HARQ feedback.

(165) Similar to the example of FIG. **8**, the UE **1004** may miss receiving a first downlink grant, may receive a second downlink grant, and may miss receiving a third downlink grant. Based on the three downlink grants transmitted by the network entity **1002**, the network entity **1002** may expect to receive first HARQ feedback **1030** (“DL Grant 1 ACK/NACK”) with a payload size of one bit at time T2, may expect to receive second HARQ feedback **1032** (“DL Grant 2 ACK/NACK”) with a payload size of two bits at time T4, and may expect to receive third HARQ feedback **1034** (“DL

Grant 3 ACK/NACK") with a payload size of three bits at time T5. However, based on the downlink grants received by the UE **1004** (e.g., the second downlink grant), the UE **1004** may determine a HARQ occasion for transmitting the second HARQ feedback **1032** at time T4.

(166) In the example of FIG. **10**, the network entity **1002** transmits an uplink grant **1010** that is received by the UE **1004** at time T1. The uplink grant **1010** may include a single uplink transmission grant with repetitions. For example, the uplink grant **1010** may schedule k repetitions of an uplink message. In the example of FIG. **10**, the uplink grant **1010** includes a repetition indicator **1012** indicating to transmit four repetitions of an uplink message. The uplink grant **1010** also includes PUSCH resources **1014** indicating time resources and/or frequency resources associated with transmitting each of the repetitions of the uplink message. For example, the PUSCH resources **1014** may allocate resources to transmit a first repetition **1020** ("PUSCH Repetition 1") at time T2, resources to transmit a second repetition **1022** ("PUSCH Repetition 2") at time T3, resources to transmit a third repetition **1024** ("PUSCH Repetition 3") at time T4, and resources to transmit a fourth repetition **1026** ("PUSCH Repetition 4") at time T5.

(167) Although the example of FIG. **10** includes the repetition indicator **1012** with the uplink grant **1010**, in other examples, the UE **1004** may receive the repetition indicator **1012** via RRC signaling, such as the example RRC signaling **436** of FIG. **4**.

(168) The uplink grant **1010** also includes a bit indicator **1016**. In the example of FIG. **10** the bit indicator **1016** is a common bit indicator indicating a quantity of x bits to multiplex on repetitions. In the example of FIG. **10**, the bit indicator **1016** indicates to multiplex 2 bits on repetitions. Aspects of the bit indicator may be implemented by an UL-TDAI parameter or a common UL-TDAI parameter.

(169) In the example of FIG. **10**, the UE **1004** is configured to determine whether to perform multiplexing on a repetition in two steps, as described in connection with the example communication flow **900** of FIG. **9**. For example, the UE **1004** may first determine if the quantity of x bits satisfies a threshold quantity of bits, as described in connection with the determination procedure **950** of FIG. **9**. For example, the UE **1004** may determine if the 2 bits indicated by the bit indicator **1016** is greater than the threshold quantity of bits. In the example of the FIG. **10**, the threshold quantity of bits is two bits, which is the same quantity of bits used by the UE **1004** to determine whether to perform multiplexing on a repetition by puncturing the repetition or by rate-matching.

(170) In the example of FIG. **10**, the UE **1004** may determine that the x bits does not satisfy the threshold quantity of bits (e.g., the 2 bits indicated by the bit indicator **1016** is not greater than the threshold quantity of bits (2 bits)). In such scenarios, the UE **1004** may then determine whether to multiplex x bits on a repetition based on whether the repetition overlaps with HARQ feedback, as described in connection with the determination procedure **960** of FIG. **9**. For example, with respect to the first repetition **1020** at time T2, the second repetition **1022** at time T3, and the fourth repetition **1026** at time T5, the UE **1004** may determine that there is no overlap with HARQ feedback. Thus, the UE **1004** may determine to transmit a first transmission **1040** at time T2, a second transmission **1042** at time T3, and a fourth transmission **1046** at time T5 based on skipping multiplexing or multiplexing zero bits on the respective repetitions, as described in connection with the second transmission **962** of FIG. **9**.

(171) In the example of FIG. **10**, the UE **1004** may determine that the third repetition **1024** at time T4 is overlapping in time with the second HARQ feedback **1032**. In such scenarios, the UE **1004** may multiplex x bits on the third repetition **1024**. For example, the UE **1004** may transmit a third transmission **1044** at time T4 including two bits multiplexed on the third repetition **1024**, as described in connection with the third transmission **966** of FIG. **9**. As described above, when the value of the x bits is less than or equal to two bits, the UE **1004** multiplexes the x bits by puncturing the repetition. For example, the UE **1004** may puncture two bits of the third repetition **1024**.

(172) Thus, as shown in FIG. **10**, the UE **1004** may multiplex {0, 0, 2, 0} bits for the repetitions of the uplink message at times T2, T3, T4, and T5, respectively. Additionally, the network entity **1002** may expect the repetitions of the uplink message to be multiplexed with expected multiplexing of {2, 0, 2, 2} bits. Although the quantity of multiplexed bits (e.g., at the UE **1004**) and the quantity of expected multiplexed bits (e.g., at the network entity **1002**) are misaligned, the impact of the mis-synchronization between the UE **1004** and the network entity **1002** is small as the network entity **1002** was expecting the first transmission **1040** at time T2 and the fourth transmission **1046** at time T5 to include the first repetition **1020** and the fourth repetition **1026**, respectively, that each include two punctured bits. However, as described above, a transmission including zero bits, one bit, or two bits punctured has little negative impact on the ability of the network entity **1002** to decode the transmission. Thus, the misalignment of the quantity of multiplexed bits (e.g., at the UE **1004**) and the quantity of expected multiplexed bits (e.g., at the network entity **1002**) may be acceptable and still provide improvements when performing HARQ feedback multiplexing on PUSCH with repetitions.
(173) FIG. **11** is a flowchart **1100** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**, and/or an apparatus **1304** of FIG. **13**). The method may facilitate higher data rates, improved capacity and/or improved spectral efficiency by enabling a UE to perform HARQ feedback multiplexing on PUSCH with repetitions based on a bit indicator included in an uplink grant.
(174) At **1102**, the UE receives an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator, as described in connection with the uplink grant **710** of FIG. **7**, the uplink grant **810** of FIG. **8**, the uplink grant **940** of FIG. **9**, and/or the uplink grant **1010** of FIG. **10**. The receiving of the uplink grant, at **1102**, may be performed by a cellular RF transceiver **1322**/the repetition component **198** of the apparatus **1304** of FIG. **13**.
(175) At **1104**, the UE transmits a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator, as described in connection with the uplink transmissions of FIG. **7**, the uplink transmissions of FIG. **8**, the uplink transmissions of FIG. **9**, and/or the uplink transmissions of FIG. **10**. The transmitting of the first repetition of the uplink transmission, at **1104**, may be performed by the cellular RF transceiver **1322**/the repetition component **198** of the apparatus **1304** of FIG. **13**.
(176) FIG. **12** is a flowchart **1200** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**, and/or an apparatus **1304** of FIG. **13**). The method may facilitate higher data rates, improved capacity and/or improved spectral efficiency by enabling a UE to perform HARQ feedback multiplexing on PUSCH with repetitions based on a bit indicator included in an uplink grant.
(177) At **1204**, the UE receives an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator, as described in connection with the uplink grant **710** of FIG. **7**, the uplink grant **810** of FIG. **8**, the uplink grant **940** of FIG. **9**, and/or the uplink grant **1010** of FIG. **10**. The receiving of the uplink grant, at **1204**, may be performed by a cellular RF transceiver **1322**/the repetition component **198** of the apparatus **1304** of FIG. **13**.
(178) At **1212**, the UE transmits a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator, as described in connection with the uplink transmissions of FIG. **7**, the uplink transmissions of FIG. **8**, the uplink transmissions of FIG. **9**, and/or the uplink transmissions of FIG. **10**. The transmitting of the first repetition of the uplink transmission, at **1212**, may be performed by the cellular RF transceiver **1322**/the repetition component **198** of the apparatus **1304** of FIG. **12**.

(179) In some examples, the downlink grant, at **1212**, may include a downlink bit indicator and the quantity of the HARQ-ACK feedback bits may be a same quantity as the bit indicator when the bit indicator is greater than or equal to the downlink bit indicator, as described in connection with Equation 3 (above). In some examples, the quantity of the HARQ-ACK feedback bits may be based on a relationship between the bit indicator and the downlink bit indicator when the downlink bit indicator is greater than the bit indicator, as described in connection with Equation 1 (above) and Equation 2 (above).
(180) In some examples, the uplink grant, at **1204**, may include an indication of the repetition quantity, as described in connection with the uplink grant **710** of FIG. **7**, the uplink grant **810** of FIG. **8**, the uplink grant **940** of FIG. **9**, and/or the uplink grant **1010** of FIG. **10**. In some examples, the UE may receive, at **1202**, an indication of the repetition quantity associated with the uplink message via RRC signaling, as described in connection with the RRC signaling **436** of FIG. **4**. The receiving of the indication, at **1202**, may be performed by the repetition component **198** of the apparatus **1304** of FIG. **13**.
(181) In some examples, the bit indicator, at **1204**, may be a first bit indicator of a bit indicator set including a number of bit indicators, as described in connection with the bit indicator set **716** of FIG. **7**. In some examples, the repetition quantity and the number of bit indicators may be a same quantity, as described in connection with the repetition indicator **712** and the bit indicator set **716** of FIG. **7**.
(182) In examples in which the bit indicator is included in a bit indicator set, the UE may, at **1206**, multiplex each repetition of the uplink transmission by a bit quantity of the HARQ-ACK feedback bits associated with a respective bit indicator of the bit indicator set, as described in connection with the multiplexed bits of the transmissions of FIG. **7**. In some examples, the multiplexing of each repetition may be based on a rate-matching. The multiplexing of each repetition of the uplink transmission, at **1206**, may be performed by the repetition component **198** of the apparatus **1304** of FIG. **13**.
(183) In some examples, the UE may multiplex, at **1208**, each repetition of the uplink transmission by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator, as described in connection with the example repetitions and the bit indicator **816** of FIG. **8**, and/or the first transmission **954** of FIG. **9**. In some examples, a second repetition of the uplink transmission may be non-overlapping in the time domain with second HARQ-ACK feedback scheduled by a second downlink grant, as described in connection with the second transmission **842** and the three multiplexed bits on the second repetition **822** of FIG. **8**. In some examples, the multiplexing of each repetition may be based on a rate-matching. The multiplexing of each repetition, at **1208**, may be performed by the repetition component **198** of the apparatus **1304** of FIG. **13**.
(184) In some examples, the UE may, at **1210**, to multiplex each repetition of the uplink transmission by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator when a value of the bit indicator is greater than a threshold, as described in connection with the determination procedure **950**, the multiplexing procedure **952**, and the first transmission **954** of FIG. **9**. In some examples, the threshold may be two bits. In some examples, a second repetition of the uplink transmission may be non-overlapping in the time domain with a second HARQ-ACK feedback scheduled by a second downlink grant, as described in connection with the second transmission **842** and the three multiplexed bits on the second repetition **822** of FIG. **8**. In some examples, the multiplexing of each repetition may be based on a rate-matching. The multiplexing of each repetition, at **1210**, may be performed by the repetition component **198** of the apparatus **1304** of FIG. **13**.
(185) In some examples, the UE may, at **1214**, receive a second uplink grant scheduling a second uplink transmission associated with a second repetition quantity, the second uplink grant including a second bit indicator. The UE may also, at **1216**, transmit a second uplink transmission repetition, the second uplink transmission repetition and second HARQ-ACK feedback scheduled by a second

downlink grant overlapping in the time domain. In some such examples, the UE may, at **1218**, skip multiplexing of a second quantity of HARQ-ACK feedback bits of the second HARQ-ACK feedback when a value of the second bit indicator is less than a threshold. In some examples, the threshold may be two bits. The receiving of the second uplink grant, at **1214**, the transmitting of the second uplink transmission repetition, at **1216**, and the skipping of the multiplexing, at **1218**, may be performed by the repetition component **198** of the apparatus **1304** of FIG. **13**.

(186) In some examples, the UE may be configured to multiplex a second repetition of the uplink transmission by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator when the second repetition is overlapping in the time domain with a second HARQ-ACK feedback scheduled by a second downlink grant and the value of the bit indicator is less than or equal to the threshold, as described in connection with the determination procedure **960**, the multiplexing procedure **964**, and the third transmission **966** of FIG. **9**. In some examples, the UE may multiplex the second repetition based on puncturing the second repetition based on the quantity of the HARQ-ACK feedback bits indicated by the bit indicator.

(187) FIG. **13** is a diagram **1300** illustrating an example of a hardware implementation for an apparatus **1304**. The apparatus **1304** may be a UE, a component of a UE, or may implement UE functionality. In some aspects, the apparatus **1304** may include a cellular baseband processor **1324** (also referred to as a modem) coupled to one or more transceivers (e.g., a cellular RF transceiver **1322**). The cellular baseband processor **1324** may include on-chip memory **1325**. In some aspects, the apparatus **1304** may further include one or more subscriber identity modules (SIM) cards **1320** and an application processor **1306** coupled to a secure digital (SD) card **1308** and a screen **1310**. The application processor **1306** may include on-chip memory **1307**. In some aspects, the apparatus **1304** may further include a Bluetooth module **1312**, a WLAN module **1314**, an SPS module **1316** (e.g., GNSS module), one or more sensor modules **1318** (e.g., barometric pressure sensor/altimeter; motion sensor such as inertial measurement unit (IMU), gyroscope, and/or accelerometer(s); light detection and ranging (LIDAR), radio assisted detection and ranging (RADAR), sound navigation and ranging (SONAR), magnetometer, audio and/or other technologies used for positioning), additional memory modules **1326**, a power supply **1330**, and/or a camera **1332**. The Bluetooth module **1312**, the WLAN module **1314**, and the SPS module **1316** may include an on-chip transceiver (TRX) (or in some cases, just a receiver (RX)). The Bluetooth module **1312**, the WLAN module **1314**, and the SPS module **1316** may include their own dedicated antennas and/or utilize one or more antennas **1380** for communication. The cellular baseband processor **1324** communicates through transceiver(s) (e.g., the cellular RF transceiver **1322**) via one or more antennas **1380** with the UE **104** and/or with an RU associated with a network entity **1302**. The cellular baseband processor **1324** and the application processor **1306** may each include a computer-readable medium/memory, such as the on-chip memory **1325**, and the on-chip memory **1307**, respectively. The additional memory modules **1326** may also be considered a computer-readable medium/memory. Each computer-readable medium/memory (e.g., the on-chip memory **1325**, the on-chip memory **1307**, and/or the additional memory modules **1326**) may be non-transitory. The cellular baseband processor **1324** and the application processor **1306** are each responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the cellular baseband processor **1324**/application processor **1306**, causes the cellular baseband processor **1324**/application processor **1306** to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the cellular baseband processor **1324**/application processor **1306** when executing software. The cellular baseband processor **1324**/application processor **1306** may be a component of the UE **350** and may include the memory **360** and/or at least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359**. In one configuration, the apparatus **1304** may be a processor chip (modem and/or application) and include just the cellular baseband processor **1324** and/or the

application processor **1306**, and in another configuration, the apparatus **1304** may be the entire UE (e.g., see the UE **350** of FIG. **3**) and include the additional modules of the apparatus **1304**.

(188) As discussed supra, the repetition component **198** is configured to receive an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator. The repetition component **198** may also be configured to transmit a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator.

(189) The repetition component **198** may be within the cellular baseband processor **1324**, the application processor **1306**, or both the cellular baseband processor **1324** and the application processor **1306**. The repetition component **198** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof.

(190) As shown, the apparatus **1304** may include a variety of components configured for various functions. For example, the repetition component **198** may include one or more hardware components that perform each of the blocks of the algorithm in the flowcharts of FIG. **11** and/or FIG. **12**.

(191) In one configuration, the apparatus **1304**, and in particular the cellular baseband processor **1324** and/or the application processor **1306**, includes means for receiving an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator. The example apparatus **1304** also includes means for transmitting a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator.

(192) In another configuration, the example apparatus **1304** also includes means for multiplexing each repetition of the uplink transmission by a bit quantity of the HARQ-ACK feedback bits associated with a respective bit indicator of the bit indicator set.

(193) In another configuration, the example apparatus **1304** also includes means for multiplexing each repetition of the uplink transmission by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator.

(194) In another configuration, the example apparatus **1304** also includes means for multiplexing each repetition of the uplink transmission by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator when a value of the bit indicator is greater than a threshold.

(195) In another configuration, the example apparatus **1304** also includes means for receiving a second uplink grant scheduling a second uplink transmission associated with a second repetition quantity, the second uplink grant including a second bit indicator. The example apparatus **1304** also includes means for transmitting a second uplink transmission repetition, the second uplink transmission repetition and second HARQ-ACK feedback scheduled by a second downlink grant overlapping in the time domain. The example apparatus **1304** also includes means for skipping multiplexing of a second quantity of bits of the second HARQ-ACK feedback when a value of the second bit indicator is less than a threshold.

(196) In another configuration, the example apparatus **1304** also includes means for receiving an indication of the repetition quantity via RRC signaling.

(197) The means may be the repetition component **198** of the apparatus **1304** configured to perform the functions recited by the means. As described supra, the apparatus **1304** may include the TX processor **368**, the RX processor **356**, and the controller/processor **359**. As such, in one configuration, the means may be the TX processor **368**, the RX processor **356**, and/or the

controller/processor **359** configured to perform the functions recited by the means.

(198) FIG. **14** is a flowchart **1400** of a method of wireless communication. The method may be performed by a network node or by one or more components of a network node (e.g., the base station **102**, the base station **310**; the CU **110**; the DU **130**; the RU **140**; and/or a network entity **1602** of FIG. **16**). The method may facilitate higher data rates, improved capacity and/or improved spectral efficiency by enabling a UE to perform HARQ feedback multiplexing on PUSCH with repetitions based on a bit indicator included in an uplink grant.

(199) At **1402**, the network node outputs an uplink grant scheduling an uplink transmission associated with a repetition quantity at a UE, the uplink grant including a bit indicator, as described in connection with the uplink grant **710** of FIG. **7**, the uplink grant **810** of FIG. **8**, the uplink grant **940** of FIG. **9**, and/or the uplink grant **1010** of FIG. **10**. In some aspects, the network node may transmit the uplink grant scheduling the uplink transmission associated with a repetition quantity at the UE. The outputting of the uplink grant, at **1402**, may be performed by the scheduling component **199** of the network entity **1602** of FIG. **16**.

(200) At **1404**, the network node obtains a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator, as described in connection with the uplink transmissions of FIG. **7**, the uplink transmissions of FIG. **8**, the uplink transmissions of FIG. **9**, and/or the uplink transmissions of FIG. **10**. In some aspects, the network node may receive the first repetition of the uplink transmission. The obtaining of the first repetition, at **1404**, may be performed by the scheduling component **199** of the network entity **1602** of FIG. **16**.

(201) FIG. **15** is a flowchart **1500** of a method of wireless communication. The method may be performed by a network node or by one or more components of a network node (e.g., the base station **102**, the base station **310**; the CU **110**; the DU **130**; the RU **140**; and/or a network entity **1602** of FIG. **16**). The method may facilitate higher data rates, improved capacity and/or improved spectral efficiency by enabling a UE to perform HARQ feedback multiplexing on PUSCH with repetitions based on a bit indicator included in an uplink grant.

(202) At **1504**, the network node outputs an uplink grant scheduling an uplink transmission associated with a repetition quantity at a UE, the uplink grant including a bit indicator, as described in connection with the uplink grant **710** of FIG. **7**, the uplink grant **810** of FIG. **8**, the uplink grant **940** of FIG. **9**, and/or the uplink grant **1010** of FIG. **10**. In some aspects, the network node may transmit the uplink grant scheduling the uplink transmission associated with a repetition quantity at the UE. The outputting of the uplink grant, at **1504**, may be performed by the scheduling component **199** of the network entity **1602** of FIG. **16**.

(203) At **1506**, the network node obtains a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator, as described in connection with the uplink transmissions of FIG. **7**, the uplink transmissions of FIG. **8**, the uplink transmissions of FIG. **9**, and/or the uplink transmissions of FIG. **10**. In some aspects, the network node may receive the first repetition of the uplink transmission. The obtaining of the first repetition, at **1506**, may be performed by the scheduling component **199** of the network entity **1602** of FIG. **16**.

(204) In some examples, the downlink grant, at **1506**, may include a downlink bit indicator and the quantity of the HARQ-ACK feedback bits may be a same quantity as the bit indicator when the bit indicator is greater than or equal to the downlink bit indicator, as described in connection with Equation 3 (above). In some examples, the quantity of the HARQ-ACK feedback bits may be based on a relationship between the bit indicator and the downlink bit indicator when the downlink

bit indicator is greater than the bit indicator, as described in connection with Equation 1 (above) and Equation 2 (above).

(205) In some examples, the uplink grant, at **1504**, may include an indication of the repetition quantity, as described in connection with the uplink grant **710** of FIG. **7**, the uplink grant **810** of FIG. **8**, the uplink grant **940** of FIG. **9**, and/or the uplink grant **1010** of FIG. **10**. In some examples, the network node may output, at **1502**, an indication of the repetition quantity associated with the uplink message via RRC signaling, as described in connection with the RRC signaling **436** of FIG. **4**. The outputting of the indication, at **1502**, may be performed by the scheduling component **199** of the network entity **1602** of FIG. **16**.

(206) In some examples, the bit indicator, at **1504**, may be a first bit indicator of a bit indicator set including a number of bit indicators, as described in connection with the bit indicator set **716** of FIG. **7**. In some examples, the repetition quantity and the number of bit indicators may be a same quantity, as described in connection with the repetition indicator **712** and the bit indicator set **716** of FIG. **7**.

(207) In examples in which the bit indicator of **1504** is included in a bit indicator set, each repetition of the uplink transmission may be multiplexed by a bit quantity of the HARQ-ACK feedback bits associated with a respective bit indicator of the bit indicator set, as described in connection with the multiplexed bits of the transmissions of FIG. **7**. In some examples, the multiplexing of each repetition may be based on a rate-matching.

(208) In some examples, each repetition of the uplink transmission, at **1506**, may be multiplexed by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator, as described in connection with the example repetitions and the bit indicator **816** of FIG. **8**, and/or the first transmission **954** of FIG. **9**. In some examples, a second repetition of the uplink transmission may be non-overlapping in the time domain with second HARQ-ACK feedback scheduled by a second downlink grant, as described in connection with the second transmission **842** and the three multiplexed bits on the second repetition **822** of FIG. **8**. In some examples, the multiplexing of each repetition may be based on a rate-matching.

(209) In some examples, each repetition of the uplink transmission, at **1506**, may be multiplexed by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator when a value of the bit indicator is greater than a threshold, as described in connection with the determination procedure **950**, the multiplexing procedure **952**, and the first transmission **954** of FIG. **9**. In some examples, the threshold may be two bits. In some examples, a second repetition of the uplink transmission may be multiplexed by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator when the second repetition of the uplink transmission is non-overlapping in the time domain with a second HARQ-ACK feedback scheduled by a second downlink grant, as described in connection with the second transmission **842** and the three multiplexed bits on the second repetition **822** of FIG. **8**. In some examples, the multiplexing of each repetition may be based on a rate-matching.

(210) In some examples, the network node may, at **1508**, output a second uplink grant scheduling a second uplink transmission associated with a second repetition quantity, the second uplink grant including a second bit indicator. As an example, the network node may transmit the second uplink grant scheduling the second uplink transmission associated with the second repetition quantity. The network node may, at **1510**, obtain a second uplink transmission repetition, the second uplink transmission repetition and second HARQ-ACK feedback scheduled by a second downlink grant overlapping in the time domain. In some such examples, the second uplink transmission repetition may exclude multiplexing of a second quantity of HARQ-ACK feedback bits of the second HARQ-ACK feedback when a value of the second bit indicator is less than a threshold. In some examples, the threshold may be two bits. The outputting of the second uplink grant, at **1508**, and the obtaining of the second uplink transmission repetition, at **1510**, may be performed by the scheduling component **199** of the network entity **1602** of FIG. **16**.

(211) In some examples, a second repetition of the uplink transmission may be multiplexed by the

quantity of the HARQ-ACK feedback bits indicated by the bit indicator when the second repetition is overlapping in the time domain with a second HARQ-ACK feedback scheduled by a second downlink grant and the value of the bit indicator is less than or equal to the threshold, as described in connection with the determination procedure **960**, the multiplexing procedure **964**, and the third transmission **966** of FIG. **9**. In some examples, the second repetition may be multiplexed based on puncturing the second repetition based on the quantity of the HARQ-ACK feedback bits indicated by the bit indicator.

(212) FIG. **16** is a diagram **1600** illustrating an example of a hardware implementation for a network entity **1602**. The network entity **1602** may be a BS, a component of a BS, or may implement BS functionality. The network entity **1602** may include at least one of a CU **1610**, a DU **1630**, or an RU **1640**. For example, depending on the layer functionality handled by the scheduling component **199**, the network entity **1602** may include the CU **1610**; both the CU **1610** and the DU **1630**; each of the CU **1610**, the DU **1630**, and the RU **1640**; the DU **1630**; both the DU **1630** and the RU **1640**; or the RU **1640**. The CU **1610** may include a CU processor **1612**. The CU processor **1612** may include on-chip memory **1613**. In some aspects, may further include additional memory modules **1614** and a communications interface **1618**. The CU **1610** communicates with the DU **1630** through a midhaul link, such as an F1 interface. The DU **1630** may include a DU processor **1632**. The DU processor **1632** may include on-chip memory **1633**. In some aspects, the DU **1630** may further include additional memory modules **1634** and a communications interface **1638**. The DU **1630** communicates with the RU **1640** through a fronthaul link. The RU **1640** may include an RU processor **1642**. The RU processor **1642** may include on-chip memory **1643**. In some aspects, the RU **1640** may further include additional memory modules **1644**, one or more transceivers **1646**, antennas **1680**, and a communications interface **1648**. The RU **1640** communicates with the UE **104**. The on-chip memories (e.g., the on-chip memory **1613**, the on-chip memory **1633**, and/or the on-chip memory **1643**) and/or the additional memory modules (e.g., the additional memory modules **1614**, the additional memory modules **1634**, and/or the additional memory modules **1644**) may each be considered a computer-readable medium/memory. Each computer-readable medium/memory may be non-transitory. Each of the CU processor **1612**, the DU processor **1632**, the RU processor **1642** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the corresponding processor(s) causes the processor(s) to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the processor(s) when executing software.

(213) As discussed supra, the scheduling component **199** is configured to output an uplink grant scheduling an uplink transmission associated with a repetition quantity at a UE, the uplink grant including a bit indicator. The scheduling component **199** may also be configured to obtain a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator.

(214) The scheduling component **199** may be within one or more processors of one or more of the CU **1610**, DU **1630**, and the RU **1640**. The scheduling component **199** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof.

(215) The network entity **1602** may include a variety of components configured for various functions. For example, the scheduling component **199** may include one or more hardware components that perform each of the blocks of the algorithm in the flowcharts of FIG. **14** and/or FIG. **15**.

(216) In one configuration, the network entity **1602** includes means for outputting an uplink grant scheduling an uplink transmission associated with a repetition quantity at a UE, the uplink grant including a bit indicator. The example network entity **1602** also includes means for obtaining a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a PDSCH scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator.

(217) In another configuration, the example network entity **1602** also includes means for outputting a second uplink grant scheduling a second uplink transmission associated with a second repetition quantity, the second uplink grant including a second bit indicator. The example network entity **1602** also includes means for obtaining a second uplink transmission repetition, the second uplink transmission and second HARQ-ACK feedback scheduled by a second downlink grant overlapping in the time domain, where the second uplink transmission repetition excludes multiplexing of a second quantity of bits of the second HARQ-ACK feedback when a value of the second bit indicator is less than a threshold.

(218) In another configuration, the example network entity **1602** also includes means for outputting an indication of the repetition quantity via RRC signaling.

(219) The means may be the scheduling component **199** of the network entity **1602** configured to perform the functions recited by the means. As described supra, the network entity **1602** may include the TX processor **316**, the RX processor **370**, and the controller/processor **375**. As such, in one configuration, the means may be the TX processor **316**, the RX processor **370**, and/or the controller/processor **375** configured to perform the functions recited by the means.

(220) Aspects disclosed herein provide techniques for improving HARQ feedback multiplexing on PUSCH with repetitions. The disclosed techniques may provide higher data rates, improve capacity, and/or improve spectral efficiency. For example, aspects disclosed herein may configure a UE to determine to perform multiplexing on a repetition regardless of whether the repetition overlaps in time with HARQ feedback. Additionally, the UE may perform the multiplexing based on a quantity of bits indicated by an uplink grant. By determining to perform the multiplexing based on the quantity of bits indicated by an uplink grant (e.g., a bit indicator) and regardless of whether there is an overlap in time, the UE may resolve concerns associated with missed downlink grants. In some examples, the bit indicator may include a bit indicator set including a number of bit indicators. For example, the quantity of bit indicators in the bit indicator set may correspond to a quantity of repetitions configured for an uplink message. The quantity of repetitions may be indicated via the uplink grant and/or via radio resource control (RRC) signaling. In some such examples, the UE may use a respective bit indicator of the bit indicator set to determine a quantity of z bits to multiplex on a respective repetition. In other examples, the bit indicator set may include one bit indicator that is applied to each repetition of the uplink message.

(221) In another aspect, the UE may be configured to determine whether to perform multiplexing on a repetition based on a bit indicator included in an uplink grant. For example, when the value of the bit indicator satisfies a threshold (e.g., x bits of the bit indicator is greater than or equal to the threshold), the UE may perform multiplexing on a repetition regardless of whether the repetition overlaps in time with HARQ feedback scheduled by a downlink grant. In examples in which the value of the bit indicator does not satisfy the threshold (e.g., the x bits is less than the threshold), then the UE may multiplex the x bits of the HARQ feedback on a repetition when the repetition overlaps in time with the HARQ feedback. The UE may also skip multiplexing or multiplex zero bits on a repetition when the repetition is non-overlapping in time with the HARQ feedback.

(222) It is understood that the specific order or hierarchy of blocks in the processes/flowcharts disclosed is an illustration of example approaches. Based upon design preferences, it is understood that the specific order or hierarchy of blocks in the processes/flowcharts may be rearranged. Further, some blocks may be combined or omitted. The accompanying method claims present

elements of the various blocks in a sample order, and are not limited to the specific order or hierarchy presented.

(223) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not limited to the aspects described herein, but are to be accorded the full scope consistent with the language claims. Reference to an element in the singular does not mean "one and only one" unless specifically so stated, but rather "one or more." Terms such as "if," "when," and "while" do not imply an immediate temporal relationship or reaction. That is, these phrases, e.g., "when," do not imply an immediate action in response to or during the occurrence of an action, but simply imply that if a condition is met then an action will occur, but without requiring a specific or immediate time constraint for the action to occur. The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any aspect described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term "some" refers to one or more. Combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" include any combination of A, B, and/or C, and may include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. Sets should be interpreted as a set of elements where the elements number one or more. Accordingly, for a set of X, X would include one or more elements. If a first apparatus receives data from or transmits data to a second apparatus, the data may be received/transmitted directly between the first and second apparatuses, or indirectly between the first and second apparatuses through a set of apparatuses. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are encompassed by the claims. Moreover, nothing disclosed herein is dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words "module," "mechanism," "element," "device," and the like may not be a substitute for the word "means." As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase "means for."

(224) As used herein, the phrase "based on" shall not be construed as a reference to a closed set of information, one or more conditions, one or more factors, or the like. In other words, the phrase "based on A" (where "A" may be information, a condition, a factor, or the like) shall be construed as "based at least on A" unless specifically recited differently.

(225) The following aspects are illustrative only and may be combined with other aspects or teachings described herein, without limitation.

(226) Aspect 1 is a method of wireless communication at a UE, comprising: receiving an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator; and transmitting a first repetition of the uplink transmission, the first repetition multiplexing a quantity of Hybrid Automatic Repeat Request (HARD) acknowledgement (HARQ-ACK) feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator.

(227) Aspect 2 is the method of aspect 1, further including that the bit indicator is a first bit indicator of a bit indicator set including a number of bit indicators, the repetition quantity and the number of bit indicators being a same quantity.

(228) Aspect 3 is the method of any of aspects 1 and 2, further including: multiplexing each repetition of the uplink transmission by a bit quantity of the HARQ-ACK feedback bits associated with a respective bit indicator of the bit indicator set.
(229) Aspect 4 is the method of any of aspects 1 to 3, further including that multiplexing of each repetition is based on a rate-matching.
(230) Aspect 5 is the method of any of aspects 1 to 4, further including: multiplexing each repetition of the uplink transmission by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator.
(231) Aspect 6 is the method of any of aspects 1 to 5, further including that a second repetition of the uplink transmission is non-overlapping in the time domain with second HARQ-ACK feedback scheduled by a second downlink grant.
(232) Aspect 7 is the method of any of aspects 1 to 6, further including that multiplexing of each repetition is based on a rate-matching.
(233) Aspect 8 is the method of any of aspects 1 to 7, further including: multiplexing each repetition of the uplink transmission by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator when a value of the bit indicator is greater than a threshold.
(234) Aspect 9 is the method of any of aspects 1 to 8, further including that the threshold is two bits.
(235) Aspect 10 is the method of any of aspects 1 to 9, further including that a second repetition of the uplink transmission is non-overlapping in the time domain with a second HARQ-ACK feedback scheduled by a second downlink grant.
(236) Aspect 11 is the method of any of aspects 1 to 10, further including that multiplexing of each repetition is based on a rate-matching.
(237) Aspect 12 is the method of any of aspects 1 to 11, further including: receiving a second uplink grant scheduling a second uplink transmission associated with a second repetition quantity, the second uplink grant including a second bit indicator; transmitting a second uplink transmission repetition, the second uplink transmission repetition and second HARQ-ACK feedback scheduled by a second downlink grant overlapping in the time domain; and skipping multiplexing of a second quantity of bits of the second HARQ-ACK feedback when a value of the second bit indicator is less than a threshold
(238) Aspect 13 is the method of any of aspects 1 to 12, further including that the threshold is two bits.
(239) Aspect 14 is the method of any of aspects 1 to 13, further including that the downlink grant includes a downlink bit indicator and the quantity of the HARQ-ACK feedback bits being a same quantity as the bit indicator when the bit indicator is greater than or equal to the downlink bit indicator.
(240) Aspect 15 is the method of any of aspects 1 to 14, further including that the quantity of the HARQ-ACK feedback bits being based on a relationship between the bit indicator and the downlink bit indicator when the downlink bit indicator is greater than the bit indicator.
(241) Aspect 16 is the method of any of aspects 1 to 15, further including that the uplink grant includes an indication of the repetition quantity.
(242) Aspect 17 is the method of any of aspects 1 to 16, further including: receiving an indication of the repetition quantity via radio resource control (RRC) signaling.
(243) Aspect 18 is an apparatus for wireless communication at a UE including at least one processor coupled to a memory and configured to implement any of aspects 1 to 17.
(244) In aspect 19, the apparatus of aspect 18 further includes at least one antenna coupled to the at least one processor.
(245) In aspect 20, the apparatus of aspect 18 or 19 further includes a transceiver coupled to the at least one processor.
(246) Aspect 21 is an apparatus for wireless communication including means for implementing any

of aspects 1 to 17.
(247) In aspect 22, the apparatus of aspect 21 further includes at least one antenna coupled to the means to perform the method of any of aspects 1 to 17.
(248) In aspect 23, the apparatus of aspect 21 or 22 further includes a transceiver coupled to the means to perform the method of any of aspects 1 to 17.
(249) Aspect 24 is a non-transitory computer-readable storage medium storing computer executable code, where the code, when executed, causes a processor to implement any of aspects 1 to 17.
(250) Aspect 25 is a method of wireless communication at a network node, comprising: outputting an uplink grant scheduling an uplink transmission associated with a repetition quantity at a user equipment (UE), the uplink grant including a bit indicator; and obtaining a first repetition of the uplink transmission, the first repetition multiplexing a quantity of Hybrid Automatic Repeat Request (HARD) acknowledgement (HARQ-ACK) feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator.
(251) Aspect 26 is the method of aspect 25, further including that the bit indicator is a first bit indicator of a bit indicator set including a number of bit indicators, the repetition quantity and the number of bit indicators being a same quantity.
(252) Aspect 27 is the method of any of aspects 25 and 26, further including that each repetition of the uplink transmission is multiplexed by a bit quantity of the HARQ-ACK feedback bits associated with a respective bit indicator of the bit indicator set.
(253) Aspect 28 is the method of any of aspects 25 to 27, further including that multiplexing of each repetition is based on a rate-matching.
(254) Aspect 29 is the method of any of aspects 25 to 28, further including that each repetition of the uplink transmission is multiplexed by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator.
(255) Aspect 30 is the method of any of aspects 25 to 29, further including that a second repetition of the uplink transmission is non-overlapping in the time domain with second HARQ-ACK feedback scheduled by a second downlink grant.
(256) Aspect 31 is the method of any of aspects 25 to 30, further including that multiplexing of each repetition is based on a rate-matching.
(257) Aspect 32 is the method of any of aspects 25 to 31, further including that each repetition of the uplink transmission is multiplexed by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator when a value of the bit indicator is greater than a threshold.
(258) Aspect 33 is the method of any of aspects 25 to 32, further including that the threshold is two bits.
(259) Aspect 34 is the method of any of aspects 25 to 33, further including that a second repetition of the uplink transmission is non-overlapping in the time domain with a second HARQ-ACK feedback scheduled by a second downlink grant.
(260) Aspect 35 is the method of any of aspects 25 to 34, further including that multiplexing of each repetition is based on a rate-matching.
(261) Aspect 36 is the method of any of aspects 25 to 35, further including: outputting a second uplink grant scheduling a second uplink transmission associated with a second repetition quantity, the second uplink grant including a second bit indicator; and obtaining a second uplink transmission repetition, the second uplink transmission and second HARQ-ACK feedback scheduled by a second downlink grant overlapping in the time domain, wherein the second uplink transmission repetition excludes multiplexing of a second quantity of bits of the second HARQ-ACK feedback when a value of the second bit indicator is less than a threshold.
(262) Aspect 37 is the method of any of aspects 25 to 36, further including that the threshold is two bits.

(263) Aspect 38 is the method of any of aspects 25 to 37, further including that the downlink grant includes a downlink bit indicator and the quantity of the HARQ-ACK feedback bits being a same quantity as the bit indicator when the bit indicator is greater than or equal to the downlink bit indicator.
(264) Aspect 39 is the method of any of aspects 25 to 38, further including that the quantity of the HARQ-ACK feedback bits being based on a relationship between the bit indicator and the downlink bit indicator when the downlink bit indicator is greater than the bit indicator.
(265) Aspect 40 is the method of any of aspects 25 to 39, further including that the uplink grant includes an indication of the repetition quantity.
(266) Aspect 41 is the method of any of aspects 25 to 40, further including: outputting an indication of the repetition quantity via radio resource control (RRC) signaling.
(267) Aspect 42 is an apparatus for wireless communication at a network node including at least one processor coupled to a memory and configured to implement any of aspects 25 to 41.
(268) In aspect 43, the apparatus of aspect 42 further includes at least one antenna coupled to the at least one processor.
(269) In aspect 44, the apparatus of aspect 42 or 43 further includes a transceiver coupled to the at least one processor.
(270) Aspect 45 is an apparatus for wireless communication including means for implementing any of aspects 25 to 41.
(271) In aspect 46, the apparatus of aspect 45 further includes at least one antenna coupled to the means to perform the method of any of aspects 25 to 41.
(272) In aspect 47, the apparatus of aspect 45 or 46 further includes a transceiver coupled to the means to perform the method of any of aspects 25 to 41.
(273) Aspect 48 is a non-transitory computer-readable storage medium storing computer executable code, where the code, when executed, causes a processor to implement any of aspects 25 to 41.

Claims

1. An apparatus for wireless communication at a user equipment (UE), comprising: memory; and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to: receive an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator set including a number of bit indicators that has a same quantity as the repetition quantity; multiplex each repetition of the uplink transmission by a bit quantity of Hybrid Automatic Repeat Request (HARQ) acknowledgement (HARQ-ACK) feedback bits associated with a respective bit indicator of the bit indicator set; and transmit a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, wherein the quantity of the HARQ-ACK feedback bits are based on a first bit indicator of the bit indicator set.
2. The apparatus of claim 1, wherein the at least one processor is further configured to: transmit each repetition of the uplink transmission, wherein each repetition includes a respective by a bit quantity of the HARQ-ACK feedback bits associated with the respective bit indicator of the bit indicator set.
3. The apparatus of claim 1, wherein the downlink grant includes a downlink bit indicator and the quantity of the HARQ-ACK feedback bits is the same quantity as the first bit indicator when the first bit indicator is greater than or equal to the downlink bit indicator.
4. The apparatus of claim 3, wherein the quantity of the HARQ-ACK feedback bits is based on a relationship between the first bit indicator and the downlink bit indicator when the downlink bit indicator is greater than the first bit indicator.

5. The apparatus of claim 1, further comprising a transceiver coupled to the at least one processor.
6. An apparatus for wireless communication at a user equipment (UE), comprising: memory; and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to: receive an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator; multiplex each repetition of the uplink transmission by a quantity of Hybrid Automatic Repeat Request (HARQ) acknowledgement (HARQ-ACK) feedback bits indicated by the bit indicator; and transmit repetitions of the uplink transmission, each repetition multiplexing the quantity of feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, wherein a first repetition and the HARQ-ACK feedback overlapping in a time domain, and wherein the quantity of the HARQ-ACK feedback bits is based on the bit indicator.
7. The apparatus of claim 6, wherein a second repetition of the uplink transmission is non-overlapping in the time domain with second HARQ-ACK feedback scheduled by a second downlink grant.
8. The apparatus of claim 6, wherein, to multiplex each repetition of the uplink transmission, the at least one processor is further configured to: multiplex each repetition of the uplink transmission by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator based on a value of the bit indicator that is greater than a threshold.
9. The apparatus of claim 8, wherein the threshold is two bits.
10. The apparatus of claim 8, wherein a second repetition of the uplink transmission is non-overlapping in the time domain with a second HARQ-ACK feedback scheduled by a second downlink grant.
11. The apparatus of claim 8, wherein the at least one processor is further configured to: receive a second uplink grant scheduling a second uplink transmission associated with a second repetition quantity, the second uplink grant including a second bit indicator; transmit a second uplink transmission repetition, the second uplink transmission repetition and second HARQ-ACK feedback scheduled by a second downlink grant overlapping in the time domain; and skip multiplexing of a second quantity of bits of the second HARQ-ACK feedback when a value of the second bit indicator is less than the threshold.
12. The apparatus of claim 6, wherein the downlink grant includes a downlink bit indicator and the quantity of the HARQ-ACK feedback bits is a same quantity as the bit indicator when the bit indicator is greater than or equal to the downlink bit indicator.
13. A method of wireless communication at a user equipment (UE), comprising: receiving an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator set including a number of bit indicators that has a same quantity as the repetition quantity; multiplexing each repetition of the uplink transmission by a bit quantity of Hybrid Automatic Repeat Request (HARQ) acknowledgement (HARQ-ACK) feedback bits associated with a respective bit indicator of the bit indicator set; and transmitting a first repetition of the uplink transmission, the first repetition multiplexing a quantity of HARQ-ACK feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, wherein the quantity of the HARQ-ACK feedback bits is based on a first bit indicator of the bit indicator set.
14. The method of claim 13, further comprising: transmitting each repetition of the uplink transmission, wherein each repetition includes a respective by a bit quantity of the HARQ-ACK feedback bits associated with the respective bit indicator of the bit indicator set.
15. A method of wireless communication at a user equipment (UE), comprising: receiving an uplink grant scheduling an uplink transmission associated with a repetition quantity, the uplink grant including a bit indicator; multiplexing each repetition of the uplink transmission by a quantity

of Hybrid Automatic Repeat Request (HARQ) acknowledgement (HARQ-ACK) feedback bits indicated by the bit indicator; and transmitting repetitions of the uplink transmission, each repetition multiplexing the quantity of feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, wherein a first repetition and the HARQ-ACK feedback overlapping in a time domain, and wherein the quantity of the HARQ-ACK feedback bits is based on the bit indicator.
16. The method of claim 15, wherein the multiplexing each repetition of the uplink transmission by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator is based on a value of the bit indicator that is greater than a threshold.
17. The method of claim 16, further comprising: receiving a second uplink grant scheduling a second uplink transmission associated with a second repetition quantity, the second uplink grant including a second bit indicator; transmitting a second uplink transmission repetition, the second uplink transmission repetition and second HARQ-ACK feedback scheduled by a second downlink grant overlapping in the time domain; and skipping multiplexing of a second quantity of bits of the second HARQ-ACK feedback when a value of the second bit indicator is less than the threshold.
18. The method of claim 15, wherein the downlink grant includes a downlink bit indicator and the quantity of the HARQ-ACK feedback bits is a same quantity as the bit indicator when the bit indicator is greater than or equal to the downlink bit indicator.
19. A method of wireless communication at a network node, comprising: transmitting an uplink grant scheduling an uplink transmission associated with a repetition quantity at a user equipment (UE), the uplink grant including a bit indicator set including a number of bit indicators that has a same quantity as the repetition quantity; and receiving a first repetition of the uplink transmission, the first repetition multiplexing a quantity of Hybrid Automatic Repeat Request (HARQ) acknowledgement (HARQ-ACK) feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, wherein the first repetition and the HARQ-ACK feedback overlapping in a time domain, and wherein the quantity of the HARQ-ACK feedback bits is based on a first bit indicator of the bit indicator set.
20. The method of claim 19, further comprising: transmitting a second uplink grant scheduling a second uplink transmission associated with a second repetition quantity, the second uplink grant including a second bit indicator; and receiving a second uplink transmission repetition, the second uplink transmission and second HARQ-ACK feedback scheduled by a second downlink grant overlapping in the time domain, wherein the second uplink transmission repetition excludes multiplexing of a second quantity of bits of the second HARQ-ACK feedback when a value of the second bit indicator is less than a threshold.
21. The method of claim 19, wherein the downlink grant includes a downlink bit indicator and the quantity of the HARQ-ACK feedback bits is the same quantity as the first bit indicator when the first bit indicator is greater than or equal to the downlink bit indicator.
22. The method of claim 21, wherein the quantity of the HARQ-ACK feedback bits is based on a relationship between the first bit indicator and the downlink bit indicator when the downlink bit indicator is greater than the first bit indicator.
23. A method of wireless communication at a network node, comprising: transmitting an uplink grant scheduling an uplink transmission associated with a repetition quantity at a user equipment (UE), the uplink grant including a bit indicator; and receiving a first repetition of the uplink transmission, the first repetition multiplexing a quantity of Hybrid Automatic Repeat Request (HARQ) acknowledgement (HARQ-ACK) feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator, wherein each repetition of the uplink transmission is multiplexed by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator.
24. An apparatus for wireless communication at a network node, comprising: memory; and at least

one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to: transmit an uplink grant scheduling an uplink transmission associated with a repetition quantity at a user equipment (UE), the uplink grant including a bit indicator set including a number of bit indicators that has a same quantity as the repetition quantity; and receive a first repetition of the uplink transmission, wherein the first repetition multiplexes a quantity of Hybrid Automatic Repeat Request (HARQ) acknowledgement (HARQ-ACK) feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, wherein the first repetition and the HARQ-ACK feedback overlap in a time domain, and wherein the quantity of the HARQ-ACK feedback bits is based on a first bit indicator of the bit indicator set.

25. The apparatus of claim 24, wherein the downlink grant includes a downlink bit indicator and the quantity of the HARQ-ACK feedback bits is the same quantity as the first bit indicator when the first bit indicator is greater than or equal to the downlink bit indicator.

26. The apparatus of claim 24, further comprising a transceiver coupled to the at least one processor.

27. An apparatus for wireless communication at a network node, comprising: a memory; and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to: transmit an uplink grant scheduling an uplink transmission associated with a repetition quantity at a user equipment (UE), the uplink grant including a bit indicator; and receive a first repetition of the uplink transmission, the first repetition multiplexing a quantity of Hybrid Automatic Repeat Request (HARQ) acknowledgement (HARQ-ACK) feedback bits of HARQ-ACK feedback for a physical downlink shared channel (PDSCH) scheduled by a downlink grant, the first repetition and the HARQ-ACK feedback overlapping in a time domain, the quantity of the HARQ-ACK feedback bits being based on the bit indicator, wherein each repetition of the uplink transmission is multiplexed by the quantity of the HARQ-ACK feedback bits indicated by the bit indicator.

28. The apparatus of claim 27, wherein the at least one processor is further configured to: transmit a second uplink grant scheduling a second uplink transmission associated with a second repetition quantity, the second uplink grant including a second bit indicator; and receive a second uplink transmission repetition, the second uplink transmission and second HARQ-ACK feedback scheduled by a second downlink grant overlapping in the time domain, wherein the second uplink transmission repetition excludes multiplexing of a second quantity of bits of the second HARQ-ACK feedback when a value of the second bit indicator is less than a threshold.